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Phase 1- Project Proposal and Planning

Phase 2 - Information System Gathering and Requirement

Phase 3 - Analysis and Design

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TABLE OF CONTENT:

1.0 Introduction.....	4
2.0 Background Study.....	6
3.0 Problem Statement.....	8
4.0 Proposed Solutions.....	10
4.1 Add-On Module: Behavior Monitoring and Emotion Tracking.....	10
4.2 Feasibility Study.....	11
4.2.1 Technical Feasibility.....	11
4.2.2 Operational Feasibility.....	11
4.2.3 Economic Feasibility (Cost-Benefit Analysis).....	12
5.0 Objectives.....	15
6.0 Scope of the Project.....	16
7.0 Project Planning.....	18
7.1 Human Resource.....	18
7.2 Work Breakdown Structure (WBS).....	20
7.3 PERT Chart (based on WBS).....	21
7.4 Gantt Chart.....	23
8.0 Benefit and Overall Summary of Proposed System.....	24
9.0 Information gathering Process.....	25
9.1 Method used.....	25
9.2 Summary from method used.....	29
10.0 : Requirement Analysis (based on AS-IS analysis).....	30
10.1 : Current business process (scenarios, workflow).....	31
10.2 : Functional Requirement (input,process,output).....	33
10.2.1 : Context Diagram.....	33
10.2.2 : Diagram 0.....	33
10.2.3 : Child Diagram.....	34
10.2.3.1: Record Daily Mood and Emotions.....	34
10.2.3.2 : Conduct Autism Screening Process.....	34
10.2.3.3: Manage Patient Data and Follow-up Scheduling.....	34
10.3 Non-functional Requirement (performance and control).....	35
10.4 Logical DFD Diagram.....	37
10.4.1 Logical Context Diagram.....	37
10.4.2 Logical Diagram 0.....	37
10.4.3.1 Record Daily Mood and Emotions.....	38
10.4.3.2 Conduct Autism Screening Process.....	39
11.0 Summary of Requirement Analysis process.....	41
12.0 System Analysis and Specification.....	43
12.1 Logical DFD To-Be System.....	43

12.1.1 Context Diagram.....	43
12.1.2 Diagram 0.....	44
12.1.3 Child Diagram.....	45
12.1.3.1 Manage Authentication and Registration.....	45
12.1.3.4 Manage Notifications and Reminder.....	48
12.1.3.5 Manage Reporting and Analytics.....	49
12.1.3.6 Manage Behavior Monitoring.....	50
12.2 Process Specification.....	50
12.2.1 Manage Authentication and Registration.....	50
Manage Authentication and Registration.....	50
Structured English.....	51
12.2.2 Manage Profile.....	52
12.2.3 Manage Screening.....	53
Structured English.....	54
12.3.4 Manage Notifications and Reminder.....	55
Structure English.....	55
12.3.5 Manage Reporting and Analytics.....	56
12.2.6 Manage Behavior Monitoring.....	58
13.1 Physical DFD To-Be System.....	59
13.1.1 Physical DFD TO-BE Diagram 0.....	59
13.1.2 Physical DFD TO-BE Child Diagram.....	60
13.1.2 .1 Manage Authentication and Registration.....	60
13.1.2 .2 Manage Profile.....	60
13.1.2 .3 Manage Screening.....	61
13.1.2 .4 Manage Notifications and Reminder.....	62
13.1.2 .5 Manage Reporting and Analytics.....	62
13.1.2 .6 Manage Behavior Monitoring.....	63
13.1.3 Physical DFD TO-BE Partitioning.....	64
13.1.4 Physical DFD TO-BE CRUD Matrix.....	70
13.1.5 Physical DFD TO-BE Event Response Table.....	71
13.1.6 Physical DFD TO-BE Structure Chart.....	72
13.1.7 Physical DFD TO-BE System Architecture.....	73
14 .0 System Wireframe.....	74
15.0 Summary of the Proposed System.....	87
16.0 References.....	88
17.0 Appendix.....	89

1.0 Introduction

The rapid rise in Autism Spectrum Disorder (ASD) diagnoses among children globally has placed great pressure on healthcare systems and families to provide early and effective intervention. Research consistently shows that a child's development is most adaptable during their first few years of life, making early detection of developmental delays critical for improving long term outcomes. While standardized clinical tools such as the Modified Checklist for Autism in Toddlers Revised (M-CHAT-R) exist, their real world application is often disturbed by uncertainties, paper based processes that lead to delayed diagnoses, lost documentation, and missed intervention opportunities.

As discussed in the Background Study, current autism screening methods in many community settings, including Malaysia, face two interconnected operational failures. First, manual paper based methods cause scoring errors and inefficient clinical workflows, delaying specialist referrals past the critical 16 to 30 month intervention window. Second, a complete absence of automated reminder systems means that follow up actions for moderate risk children or rescreening for low risk toddlers are often forgotten or ignored, resulting in children falling through the cracks. The Problem Statement further breaks these issues into three specific areas, operational inefficiencies and diagnostic delays in manual methods, data fragmentation and lack of standard regulatory profiles, and unclear follow up mechanisms with no proactive reminder system.

To directly address these gaps, this project proposes the development of an Autism Early Health Screening Support System, an integrated, cross platform solution comprising both a web based dashboard for NGOs and a mobile application for parents. Unlike existing apps that only digitize the test itself, this proposed system introduces three key innovations: an automated M-CHAT-R scoring engine that provides instant risk classification (Low, Moderate, High), a profile management module aligned with BPPOKU Part A standards to eliminate redundant data entry, and a proactive notification engine using push notifications and emails to ensure timely follow up screenings. Additionally, a Behavior Monitoring and Emotion Tracking add on module will be implemented, allowing parents to record daily moods, sleep patterns, and social interactions, generating visual trend reports that can be shared with therapists and doctors.

The feasibility of this project has been analyzed across technical, operational, and economic dimensions. Technologically, the system is achievable using widely available cross platform development tools, and the team possesses the necessary programming knowledge. Operationally, the user-friendly design meets the needs of parents and NGO administrators who are already familiar with mobile health applications. Economically, a cost benefit analysis over three years demonstrates a positive return on investment, with significant savings from reduced paperwork, minimized manual errors, and improved communication efficiency between parents and support organizations.

Therefore, the main goal of this project is to model, design, and develop a realistic prototype that transforms autism screening from an isolated paper form into a connected, digital lifecycle. The specific objectives are to build an automated M-CHAT-R screening module for real time risk scoring, integrate a BPPOKU aligned profile management system, implement an automated reminder engine for up to date actions and develop a behavior monitoring module for continuous emotional and developmental tracking. The scope of this project includes the complete development of these modules for two user groups (parents and NGO admins), delivered within a structured 12 week timeline as outlined in the project planning section.

Ultimately, this proposed system will provide parents with an accessible, all in one platform to complete screenings, track behavioral patterns, and share standardized reports with specialists, while equipping NGOs with centralized dashboards to manage support efficiently. By replacing fragmented manual workflows with an integrated digital health framework, this project aims to ensure that no child misses the critical window for early autism intervention due to preventable administrative failures.

2.0 Background Study

Autism Spectrum Disorder (ASD) is a lifelong developmental condition that affects how a person communicates, interacts with others, and experiences the world. Globally, the number of children that have been diagnosed with autism is rising rapidly. Because a child's brain develops fastest during their few years of life, catching developmental delays early is incredibly important. Early screening opens a vital window where children can get the right therapy and educational support, which drastically improves their long term growth and independence.

Medical professionals agree that using official standardized checklists is the best way to catch autism early. One of the most trusted tools for autism early screening in the world is the M-CHAT-R (Modified Checklist for Autism in Toddlers-Revised).

This checklist asks parents a simple Yes or No questions about their children behavior. For example, whether their child looks when they point at a toy or responds to their own name. Based on the answers, an official scoring system places the child into one of three risk levels which is Low Risk (Score 0-2), Moderate Risk (Score 3-7) and High Risk (Score 8-20). For low risk score, there is no immediate action is required unless new symptoms appear later. If the total scores showing for moderate risk then the child needs a second-stage follow-up test to double check the results while if the total scores showing for high risk then the child can skip the follow-up session and is referred immediately to a specialist for a formal diagnosis and fast-tracked therapy.

Even though tools like M-CHAT-R work perfectly on the paper, the actual process of getting a child screened in the real world is hard. Current methods actually face major difficulties including too much paperwork as most screenings are still done manually on paper, This leads to slow processing times, lost of documents, and human mistakes when calculating the total scores. Besides, many families do not know these tools exist or cannot easily get them from local clinics is one of the challenges also. Not to mention that, there are disconnected systems between parents, community non-profit organizations and government welfare or education departments as they do not share data smoothly. Plus, there is no daily support provided because the existing apps only focus on the test itself. They do not help parents manage their child's daily routines or provide visual communication tools while they wait for months for an official hospital appointment.

To fix these gaps, this project uses an integrated digital health framework, instead of treating autism screening as an isolated paper form, this framework turns it into a connected, digital lifecycle. By launching both a mobile app and a web platform, the framework connects all stakeholders. It digitalizes the M-CHAT-R logic to automate scoring so parents get instant, reliable results. Additionally, it links the user's profile directly to the official government welfare structures, specifically matching the Malaysian Part A of BPPOKU (Amendment 2019) standards to make official registration faster.

Building this digital support system brings plenty of benefits to the community such as specifically for parents, it puts a validated screening tool right on their smartphones. They get instant data-driven guidance without waiting for a clinic visit. For NGOs, it gives them a centralized dashboard with data analytics. This helps them see community trends, track which families need the most help, and manage support efficiently. As for schools and clinics, it helps them because the system collects official bio data based on the BPPOKU form, so parents can easily export and share accurate digital records with doctors and school boards. This cuts down administrative wait times for special education placement and healthcare aid.

While early intervention saves lives, the current reality for many families is defined by messy paperwork, delayed doctor referrals, and missed opportunities. The main challenge is not inventing a new medical test but building the modern digital infrastructure needed to make screening fast, accessible and connected. This project leads directly into our detailed problem statement and design requirements.

3.0 Problem Statement

The development of an early autism symptom detection system is being carried out to address the systematic operational problems that occur in the current autism registration space. This problem space is divided into 3 operations that are interconnected with each other.

3.1 : Operational Inefficiencies and Diagnostic Delays in Manual Methods

In traditional autism screening methods, the use of physical assessment instruments and paper-based forms such as the Modified Checklist for Autism in Toddlers, Revised (M-CHAT-R) is still widely used. During the scoring process carried out by their parents, human errors can occur when small developmental signs are not properly examined or not accurately counted. In addition, the clinical workflow becomes less organized due to the lack of an automated scoring system. Referrals to specialists are also often delayed, which can lead to longer waiting times. As a result, the important early intervention period during the age of 16 to 30 months may not be provided to children at risk of autism. Therefore, their neurodevelopment and long-term prognosis may be affected.

3.2 : Data Fragmentation and Lack of Standard Regulatory Profiles

Currently, significant differences can still be seen between the initial screening process at the community level and the formal registration process of individuals with special needs under welfare bodies. Information management is also carried out separately, where parental data collected during initial observation is not standardized or matched according to the requirements of the legal institutional framework, especially the biodata of Part A in the BPPOKU form (Amendment 2019) issued by the Department of Social Welfare. Due to the absence of a standardized data schema, manual data re-entry is still required even if the child has been suspected of being at high risk. In addition, there is also a problem faced by Non-Governmental Organizations acting as system owners because standardized profile tracking cannot be done properly. As a result, the process of verifying user background, coordinating community assistance, and maintaining auditable data records becomes more difficult to manage.

3.3 Unclear Mechanism of Follow-up Actions and Absence of Proactive Reminder System (Automated Reminders)

The Modified Checklist for Autism in Toddlers instrument used during child development screenings is not a static procedure performed only once. Based on clinical parameters, this instrument is highly related to the sensitive timeline of child development, particularly between the ages of 16 and 30 months. In addition, different follow-up actions also need to be taken based on the level of risk scores obtained during the screening process.

For example, if a child is identified as being in the “Moderate Risk” category (score 3–7), a second-level screening using the M-CHAT-R/F follow-up form should be conducted immediately to ensure that the symptoms involved can be confirmed. Meanwhile, for children in the “Low Risk” category (score 0–2) who are under 24 months of age, re-screening should still be scheduled when they reach 2 years of age to ensure that late-onset flags are not missed. However, this entire important follow-up process is often disrupted due to the lack of a digital system that can automate scheduling and periodic monitoring.

From a parent’s perspective, lack of clinical knowledge about autism procedures is considered one of the main barriers. After manual screening results are received, parents are given physical documentation, and clearly outlined follow-up screening times are often not provided. In addition, factors such as anxiety, denial, busy schedules, and lack of awareness of the importance of regular monitoring can cause early screening results to be shelved, ignored, or forgotten. Without automated reminder systems such as push notifications, automated emails, or periodic alerts sent directly to the user’s device, opportunities to detect developmental changes in children during critical developmental periods can be missed due to unintentional negligence.

4.0 Proposed Solutions

The development of an Autism Early Health Screening Support System is proposed to address the issues faced in the current autism screening and monitoring process. The blueprint specifies a web-based, mobile-responsive user interface design that allows parents and caregivers to perform digital screening while interacting with a continuous behavioral and emotional tracking layout.

The system design integrates the logical framework of the Modified Checklist for Autism in Toddlers-Revised (M-CHAT-R) instrument alongside a data schema for continuous behavioral monitoring. This replaces physical, paper-based forms with an organized, relational data structure and an intuitive user-facing presentation layer. The design also incorporates a dedicated administrative dashboard layout for non-governmental organizations (NGOs) to manage user profiles, view relational screening records, and track follow-up monitoring data.

4.1 Add-On Module: Behavior Monitoring and Emotion Tracking

The Behavior Monitoring and Emotion Tracking Module is designed as a core functional component to structurally support continuous developmental monitoring processes for children with autism.

This module is designed to help parents and caregivers record and monitor the child's daily emotions, behaviors, communication patterns, and social interactions. Children with autism often display emotional and behavioral changes that require consistent observation over time. Therefore, this module provides a centralized platform for parents to track developmental progress more effectively.

Key functions design for this Behavior Monitoring and Emotion Tracking Module include:

- Customized input forms for recording daily moods and emotional states
- Database tables structured to log repetitive behaviors and social interaction variables
- Form designs and data fields optimized for logging sleeping schedules and daily routines
- Representing behavioral trend reports and progress summaries
- Allowing parents to retrieve and review previous emotional and behavioral records
- Providing historical behavioral data for therapists and specialists

By including this module, the proposed design not only supports early autism screening but also provides continuous developmental and emotional behavior monitoring for children with autism.

4.2 Feasibility Study

A feasibility study was conducted to determine whether the proposed system design is logically practical, scalable, and compatible with modern user experiences. Three main feasibility aspects were evaluated, namely technical feasibility, operational feasibility, and economic feasibility.

4.2.1 Technical Feasibility

The proposed Autism Early Health Screening Support System is technically feasible because the interface layouts, database models, and architectural frameworks align with widely supported, modern cross-platform development standards.

The system will be developed as both a web-based and mobile-friendly platform to ensure accessibility for parents, caregivers, NGO administrators, and healthcare-related users. In addition, the development process is considered achievable because the project team possesses basic programming knowledge and technical skills required for system development. With proper project planning, task allocation, and time management, the system is expected to be completed within the allocated semester timeline.

The implementation of the M-CHAT-R screening instrument is also technically achievable because the questionnaire mainly involves structured Yes-or-No responses with predefined scoring rules. Automated scoring algorithms can therefore be programmed efficiently to calculate Low Risk, Moderate Risk, and High Risk classifications instantly after screening completion.

4.2.2 Operational Feasibility

The proposed design is operationally feasible because it directly addresses the needs of parents, caregivers, and NGO administrators involved in autism screening and developmental monitoring processes without requiring complex behavioral adjustments..

Nowadays, many parents are already familiar with using mobile applications and websites for healthcare-related services and information. Therefore, the proposed design is simple, clean, and user-friendly interface that allows users to perform autism screening and behavior monitoring activities easily. Clear instructions and straightforward navigation are also provided to ensure that users can operate the system without requiring special technical training.

The Behavior Monitoring and Emotion Tracking Module further improves the practicality of the system by allowing parents to continuously record their child's emotions, behaviors, and developmental patterns in a more organized manner. This helps parents monitor their child's progress more consistently while reducing dependency on manual observation records.

In addition, NGO administrators can monitor screening trends, behavioral records, and user activities through the centralized dashboard provided by the system. This enables organizations to identify children who may require further professional evaluation or early intervention support more efficiently.

4.2.3 Economic Feasibility (Cost-Benefit Analysis)

An economic feasibility study was conducted to identify whether the financial investment required for the proposed design justified the expected benefits. This analysis includes the calculation of present value (PV) and profitability index over a three year period.

<u>Development Cost</u>	
System Development	RM 10000
Domain Name Registration	RM 150
Screening Tool License	RM 3000
NGO staff's training	RM1000
Web hosting (first year)	RM 1300
Testing & Quality Assurance	RM1265

<u>Benefits</u>	
Paper and printing cost Reduction	RM 6000/year
Storage and space	RM 3000/year
Communication efficiency (Parents-NGO)	RM 4000/year
Error and rework reduction	RM 2000/year
Data and Report Generation efficiency	RM 4000/year

Production Cost

Maintenance and security	RM 2000/year
IS Support	RM 2000/year
Renewal for web hosting	RM 1200/year

Assumptions

Discount rate	10%
Sensitivity factor (cost)	0.7
Sensitivity factor (benefit)	0.9
Annual change in operational cost	5%
Annual change in benefit	7%

Item	Year 0	Year 1	Year 2	Year 3
Development Cost				
System Development	7 000			
Domain Name Registration	105			
Screening Tool License	2 100			
NGO staff's training	700			
Web hosting (first year)	910			
Testing & Quality Assurance	885.50			
Total	11 700.50			
Production Cost				
Maintenance and security		1 400	1 470	1 543.50
IS Support		1 400	1 470	1 543.50
Renewal for web hosting		840	882	926.10
Total		3 640	3 882	4 013.10
Accumulated Prod. Cost (Present Value)		3 309.09	3 208.26	3 015.10

Accumulated Cost		15 009.59	18 217.85	21 232.95
Benefit				
Paper and printing cost Reduction		5 400	5 778	6 182.46
Storage and space		2 700	2 889	3 091.23
Communication efficiency (Parents-NGO)		3 600	3 852	4 121.64
Error and rework reduction		1 800	1 926	2 060.82
Data and Report Generation efficiency		3 600	3 852	4 121.64
Total		17 100	18 297	19 577.79
Accumulated Benefit (Present Value)		15 545.45	15 121.49	14 694.06
Accumulated Benefit		15 545.45	30 666.94	45 361
Gain or Loss		535.86	12 449.09	24 128.05
Profitability Index	$24\,128.05 / 11\,700.50 = 2.06$			

Based on the cost-benefit analysis conducted over three years with a discount rate of 10% and sensitivity factors applied (cost x 0.7, benefit x 0.9), the proposed Autism Early Screening The Support System is economically feasible. The profitability index of 2.06 (greater than 1) confirms that the benefits significantly outweigh the costs.

5.0 Objectives

5.1: Data collection through google form

The first objective of this project is to collect raw data directly from stakeholders such as parents, guardians and teachers of children with autism. The initiative we used is a digital questionnaire via Google Form. This data collection is very important to explore the real challenges faced in the traditional screening process which is often slow, manual and unintegrated.

5.2: Data Analysis

The data obtained during the previous phase was used to systematically analyze the data to translate it into system requirements. Quantitative analysis methods were used such as pie charts and simple percentages to assess the frequency of screening delay issues faced by the community, as well as qualitative analysis used to examine written feedback on additional important features such as daily routine management or behavior monitoring. The results of the analysis can help us design the system's functional priorities.

5.3: Design website prototype (UI/UX)

Our third objective is to design a high-fidelity visual prototype that focuses on accessibility and inclusive user experience (UI/UX) for parents or guardians with autism. Through this project, we will design an interface (mockup) and a workflow simulation. However, this design prototype will plan the visual layout of all mandatory modules in detail. This includes a graphic sketch of a secure login and registration screen, a digital replica of the biodata form layout, an interface display flow for the 20 M-CHAT-R screening questions along with a simulated visualization of the automated scoring logic (Low, Medium, or High Risk), and an analytics dashboard design.

Overall, a profound impact on learning outcomes was achieved by us as System Analysis and Design students throughout the implementation of this process. Through this initial planning phase, using industry tools such as GitHub and Kanban boards was able to increase our skills in managing projects collaboratively and effectively, as well as successfully honing our critical problem-solving skills by translating real-world issues into technical solutions. The process of connecting data from the Google Form phase to the system design phase has realized academic theory and industry practice, and shaped us into future data engineers who are technically proficient and also have high empathy for the needs of end users.

6.0 Scope of the Project

In this project, we plan to design six use cases

1. Manage Authentication and Registration

- Implement a safe and secure user registration form that allows users to create an account by providing basic information such as name, email, password and phone number during the registration phase.
- Build a sophisticated authentication system that features a secure login and logout system and password recovery tools to ensure user data is protected from external access.
- Actor involved: Parent, Admin

2. Manage Profile

- Implement a comprehensive profile management module that is built entirely based on the layout of the BPPOKU Jabatan Kebajikan Masyarakat form (Pindaan 2019) Part A.
- Compile and collect user biodata for the system database, including the child's full name, identification number, birth certificate and family medical background.
- Design a centralized interface that allows parents to manage, update and review their children's personal records and progress at any time.
- Actor involved: Parent, Admin

3. Manage Screening

- Build an initial assessment module that uses the M-CHAT-R instrument.
- Provides a user-friendly interface to answer questions (yes/no) related to children's social development and behavior.
- Develop an automated scoring algorithm to determine risk levels which is Low Risk, Medium Risk and High Risk to support further diagnostic referral processes.
- Actor involved: Parent, Admin

4. Manage Notification and Reminder

- Provide automatic alert modules to ensure users receive important information on time.
- Develop automated reminder mechanisms to support follow-up, including notifying parents when re-screening is required for children under 24 months of age.
- Actor involved: Parent, Admin

5. Manage Reporting and Analytics

- Design reporting and analytics modules to provide comprehensive data-based information for further assessment and planning purposes.
- Develop a centralized platform system that allows administrators to monitor screening results and manage diagnostic referrals more systematically.
- Actor involved: Admin

6. Manage Behavior Monitoring

- Build a specific behavioral monitoring module to assess children's emotions and moods.
- Allows behaviour data to be collected consistently to support effective early intervention actions.
- Actor involved: Parent, Child

7.0 Project Planning

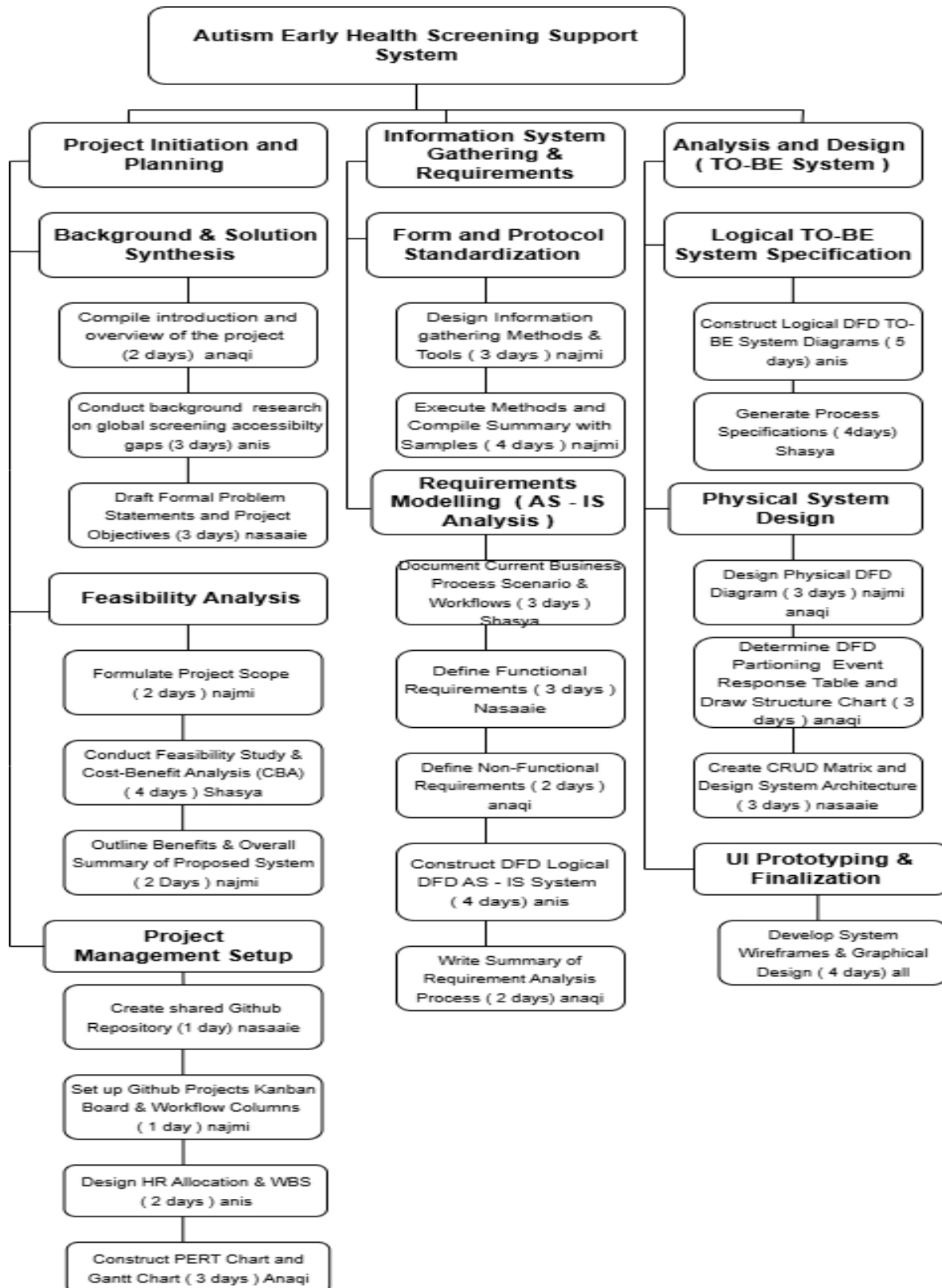
7.1 Human Resource

To keep our development organized , our team follows the first four phases of the System Developments Life Cycle (SDLC).This includes finding the main problems,gathering requirements,analyzing the system needs and creating the final design.To make sure this project work successfully , every team member works across all four SDLC phases .While each member takes charge of a specific module, everyone helps with brainstorming problems, collecting user needs,analysing data,and designing the final layouts for every part of the project.

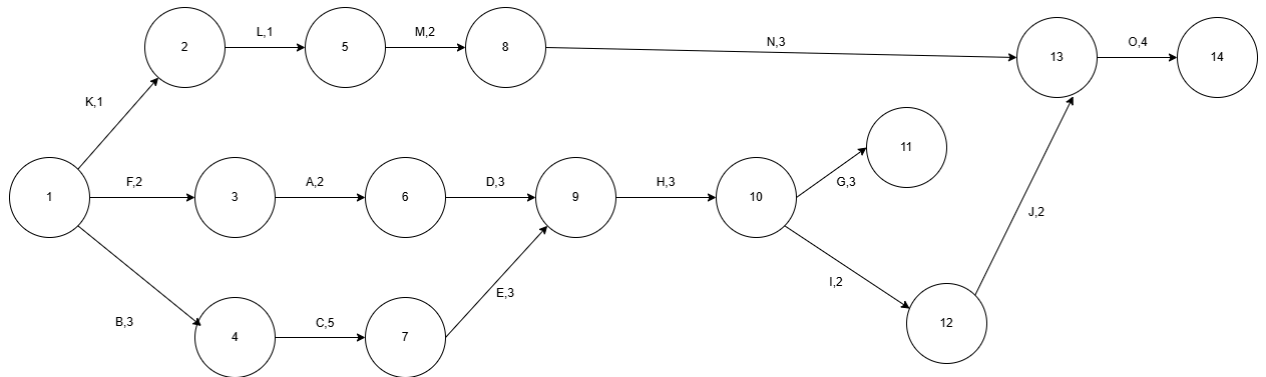
Roles	Team Members	Role responsibility
Project Manager - Timeline management - Progress monitoring	1. Muhammad Najmi Shahmi Bin Mohd Latpi	Plans the overall project schedule, checks everyday progress,updates the team GitHub repository and run team meetings to keep all SDLC phases on track.
Authentication & Registration Module	1. Nasaaie Bin Noriskamar	Leads the problem solving, requirement gathering, analysis and interface design for secure user signup,login and logout features.
User Profile Management Module	1. Anaqi Harraz Bin Mohd Azad	Leads the research,analysis and layout design for the profile forms to ensure the form captures all official data needed for BPPOKU Pindaan 2019 (Bahagian A).
Autism Screening Module (M-CHAT-R)	1. Muhammad Najmi Shahmi Bin Mohd Latpi	Leads the analysis design of the M-CHAT-R digital quiz,including the scoring logic for Low,Medium,High risk tiers.
Notification & Reminder Module	1. Shasya Shafieqah Binti Shahrudin	Leads the requirement gathering and design for the alert system, making sure users automatically get screening results and reminders.

Reporting and Analytics Module	1. Puteri Anis Annisa binti Mat Lazim	Leads the system analysis and visual design for the main dashboard , creating clear charts and summary reports for screening data.
Behavior Monitoring Module	1. Puteri Anis Annisa binti Mat Lazim 2. Shasya Shafieqah Binti Shahrudin 3. Anaqi Harraz Bin Mohd Azad	Leads the analysis and design of the mood and emotion tracker, making simple pages for users to record and see daily behavior trends.
Documentation & Planning	1. Muhammad Najmi Shahmi Bin Mohd Latpi 2. Nasaie Bin Noriskamar 3. Puteri Anis Annisa binti Mat Lazim 4. Shasya Shafieqah Binti Shahrudin 5. Anaqi Harraz Bin Mohd Azad	Every member works together during all the SDLC phases to write, review and organize project reports. The entire team helps create and check the main projects diagram including the WBS,PERT chart,Gantt chart and Data Flow Diagram to make sure everything is complete, correct and aligned with each other.

7.2 Work Breakdown Structure (WBS)



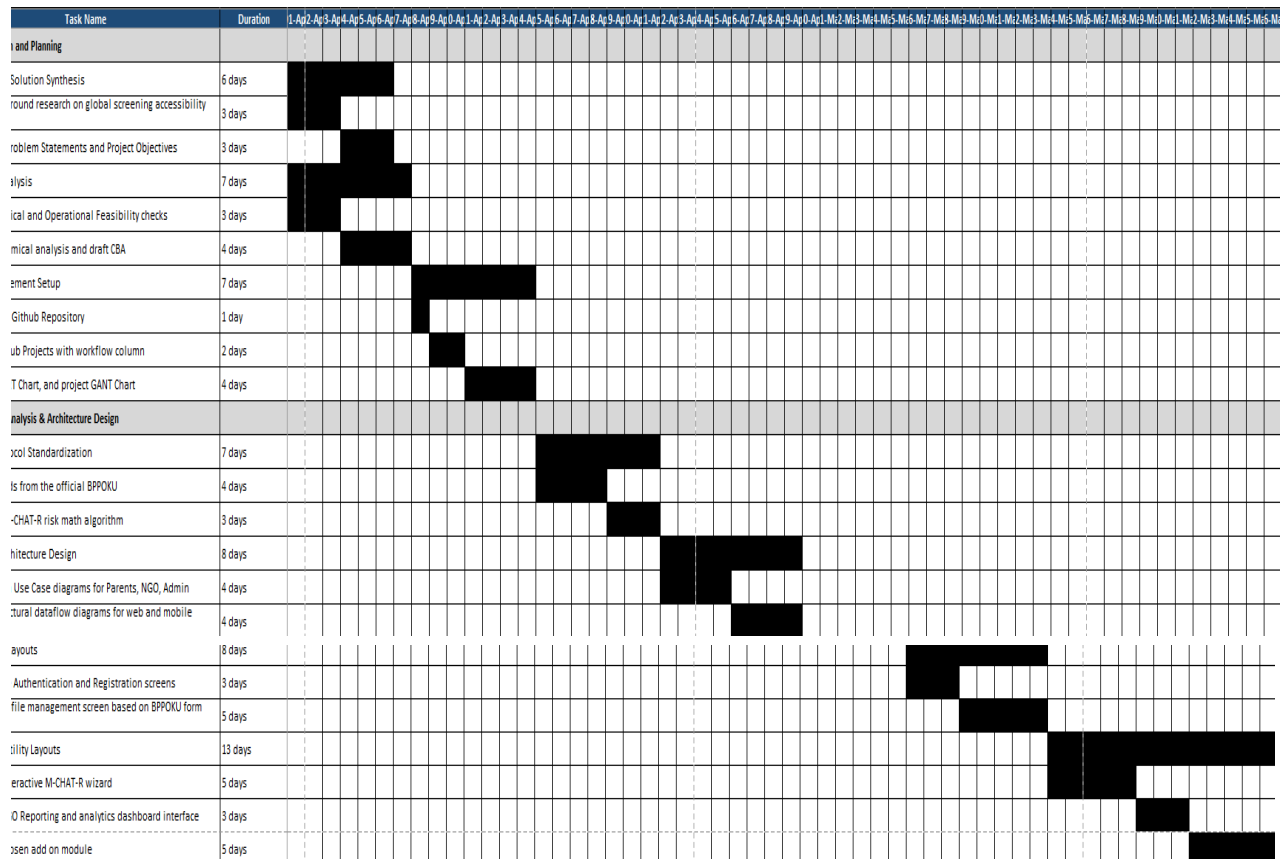
7.3 PERT Chart (based on WBS)



ID	Task Name	Duration (days)
A	Compile introduction and overview of the project	2
B	Design information gathering Methods & Tools	3
C	Construct Logical DFD TO-BE System Diagrams	5
D	Document Current Business Process Scenario & Workflows	3
E	Design Physical DFD Diagram	3
F	Formulate Project Scope	2
G	Determine DFD Partitioning Event Response Table and Draw Structure Chart	3
H	Conduct Feasibility Study & Cost-Benefit Analysis (CBA)	4
I	Outline Benefits & Overall Summary of Proposed System	2

J	Write Summary of Requirement Analysis Process	2
K	Create shared Github Repository	1
L	Set up Github Projects Kanban Board & Workflow Columns	1
M	Design HR Allocation & WBS	2
N	Construct PERT Chart and Gantt Chart	3
O	Develop System Wireframes & Graphical Design	4

7.4GanttChart



<https://docs.google.com/spreadsheets/d/1ck-cwKxj-hDn-v48vYaluse9pBMyKPmrQGHDKsfCfRE/edit?usp=sharing>

8.0 Benefit and Overall Summary of Proposed System

- Help to track child's behavioral and emotional development

The system we propose could provide a platform for parents and children to track their child's emotional states through the behavior monitoring module, all the recorded moods and behaviour patterns will be saved securely in the database. Parents can track their child's long-term developmental progress and view historical trend charts anytime and anywhere by just signing into their respective accounts.

- Provide an easier way for parents to complete early autism screening

Mentioning the section to perform childhood screening, parents will face difficulty or confusion when dealing with manual medical checklists and fragmented paperwork. This situation happens because traditional methods require extensive manual effort, leading to delayed diagnoses or lost information. Hence, when parents use our system, the standardized M-CHAT-R questionnaire is completely online with simple closed-ended questions, and the child's personal records are perfectly organized based on the official BPPOKU (Pindaan 2019) Bahagian A template. Thus, this system can help parents to save their time and handle stress perfectly through an all-in-one platform.

- Improve children's access to early clinical intervention

The proposed system increases opportunities for early treatment by using an automated scoring to quickly identify children who may be at risk. There is a feature included in our system where the program will instantly calculate answers and categorizes risks into low, medium, or high tiers, paired with automated follow-up reminders so that critical diagnostic deadlines are never missed. In addition, all information records will be systematically evaluated through an analytics dashboard by admin, improving the efficiency of tracking diagnostic referrals to healthcare providers.

In a nutshell, our proposed system could help parents to auto calculate autism risk levels and continuously monitor behaviour trends through their efforts of answering digital questionnaires and logging daily moods. All profiles and screening results will be managed securely by the administrators of the non-profit organization (NGO). Hence, what parents have to do is just to fill out the screening questions, monitor their child's emotional updates. The result also can be shared with clinical specialists to enable prompt early support and timely intervention.

9.0 Information gathering Process

The information gathering process is a key step in system analysis and design because it provides valuable insights into user needs, existing practices, and system specifications for the proposed solution. For this project, the information gathering process was conducted to identify the needs for a Behavior Monitoring Module (Emotion or Mood Tracking) in the Autism Early Screening Support System. This module was developed to assist parents in continuously recording and monitoring the emotional and behavioral patterns of autistic children. The data and feedback obtained from respondents will be used to determine functional and non-functional requirements, which will then guide the design and development of the proposed system.

9.1 Method used

To effectively collect relevant and reliable data, the main technique we used to collect data was the questionnaire method. A set of structured questionnaires was developed and distributed online to obtain the views, experiences and suggestions of individuals directly involved in the care and support of children with autism. This method was chosen because it allows data to be collected from many respondents in a short period of time. In addition, this approach facilitates the data analysis process because the information obtained is structured and systematic.

In the process of collecting information, we used Google Forms to facilitate the process of preparing, distributing and collecting data. Respondents were invited to answer the questions that we had created through an online link shared via social media. After this questionnaire was distributed, we obtained 16 respondents, the majority of whom were parents. Their responses greatly helped us in implementing the next steps.

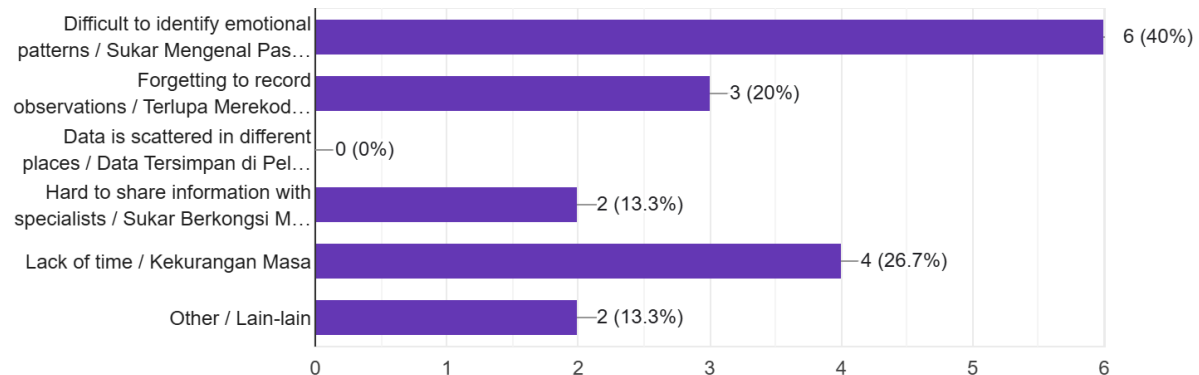
Among the 5 important questions out of the 20 questions asked are:

6. What challenges do you face when monitoring emotions or behavior? /

[Copy chart](#)

Apakah cabaran yang anda hadapi semasa memantau emosi atau tingkah laku?

15 responses

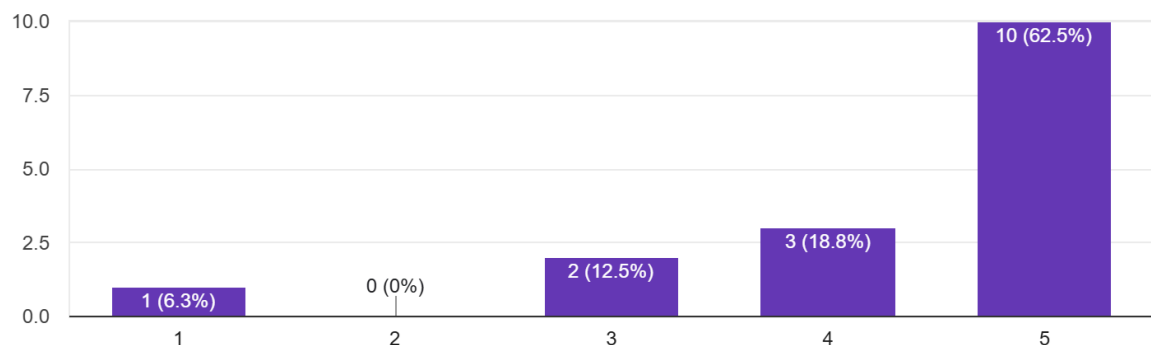


Based on the data gathered from the participants in the questionnaire , the majority of them (40%) agree it is hard to identify emotional patterns of their child. This situation illustrates that they need a system that can make it easier for them to monitor their children's emotions and behavior more accurately and systematically.

7. How important is emotion or mood tracking in supporting children with autism? / Sejauh manakah pentingnya penjejakan emosi atau mood dalam menyokong kanak-kanak autisme?

[Copy chart](#)

16 responses

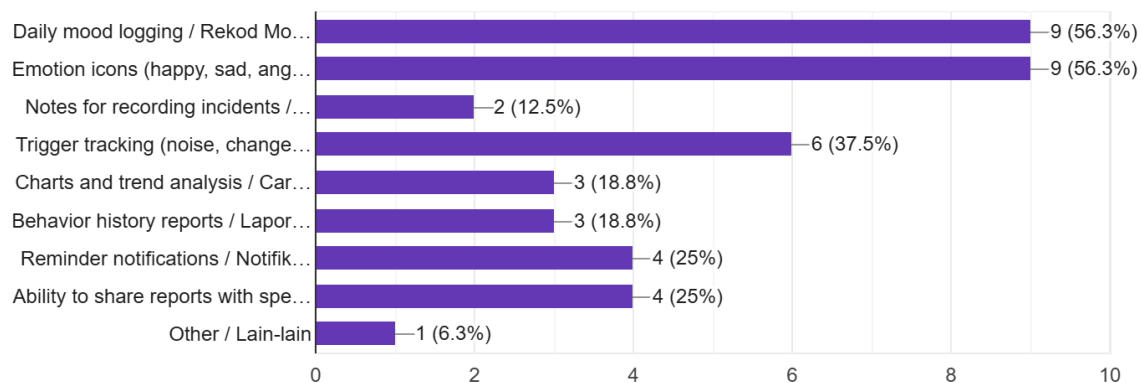


Based on the pie chart obtained, it can be concluded that parents have a high need for a system that can track their children's emotions and moods. The implementation of this system can provide parents with early information about their children's emotional and behavioral patterns, thus allowing them to take appropriate follow-up actions and obtain early intervention at the right time.

10. Which features would you like to see in the Behavior Monitoring module? /
Ciri-ciri apakah yang anda ingin lihat dalam modul Pemantauan Tingkah Laku?

 [Copy chart](#)

16 responses

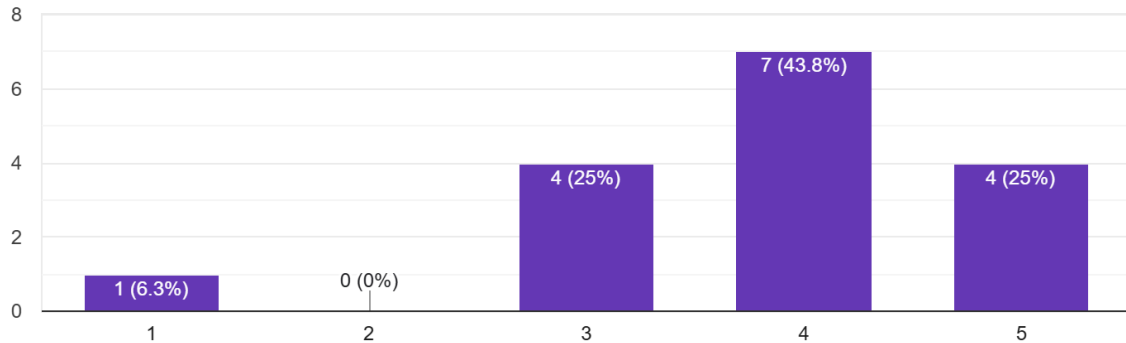


Based on the bar chart for question 10, the features that most respondents liked the most were daily mood logging and emotion icons. Each of which was chosen by 9 respondents, indicating that users really liked the easy way to record their children's daily moods. Overall, users preferred practical and user-friendly features that support daily emotion tracking.

13. How useful would visual charts showing mood trends over time be? / Sejauh manakah bergunanya carta visual yang menunjukkan trend mood dari semasa ke semasa?

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16 responses

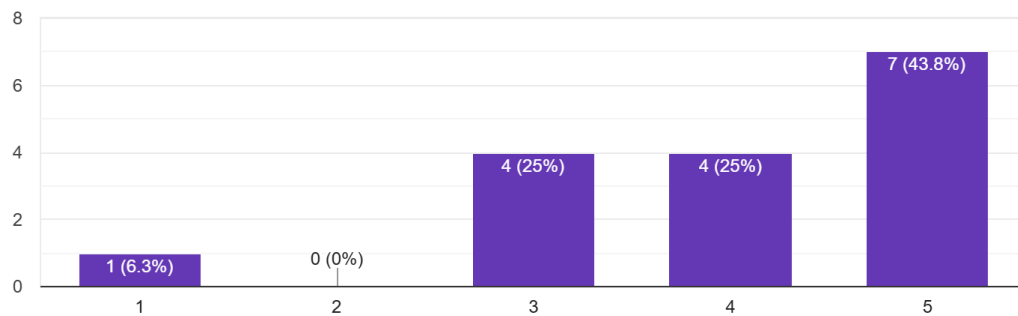


Based on the data obtained through the bar chart, the majority of respondents agreed that a visual chart that displays mood trends over time would be very useful. This can be seen when they rated 4 (43.8%) and 5 (25.0%). This finding indicates that most respondents believe that the value of visual charts can help them monitor their children's emotional patterns more effectively.

18. How concerned are you about the privacy and confidentiality of behavioral data? / Sejauh manakah anda mengambil berat tentang privasi dan kerahsiaan data tingkah laku?

 [Copy chart](#)

16 responses



Based on data from Google Forms, the majority of respondents expressed their level of concern about the privacy and confidentiality of their children's behavioral data. The highest rating was 7 out of 16 respondents (43.8%). This data shows that privacy and confidentiality of data are important aspects that must be protected.

9.2 Summary from method used

We used the questionnaire method as an information gathering process to gather views, user preferences and challenges related to the Behavior Monitoring Module. Through this method, it allows us to obtain data from individuals involved in supporting children with autism. Through the questionnaire, we can identify the current methods used to monitor children's emotions and behavior including the difficulties encountered.

In conclusion, the Google Form method allows us to collect complete feedback that includes current practices, maintaining the system, desired features and others. In addition, this method provides a systematic and efficient approach to collecting the necessary information. The responses obtained also offer an understanding of the needs and priorities required by a user. In addition, the responses obtained can also ensure that our project runs smoothly and orderly without any shortcomings.

10.0 : Requirement Analysis (based on AS-IS analysis)

Problem identification:

- **Lack of regularity in the manual screening process:** Manual assessments and physical paper forms become a reliance on existing autism screening workflows, causing the process to be slow and resource-intensive for families and healthcare providers.
- **Delays in diagnosis and a less organized follow-up schedule:**Delayed important referrals and inconsistent follow-up are often the result of traditional screening methods. The support for children who probably have a chance to be saved is indirectly lesser. Especially during the critical early developmental and therapeutic stages of their most formative years.
- **Lack of a centralized data management and tracking platform:** Managing the results of screening, managing data of patients, and monitoring long-term behavioral patterns is becoming less efficient due to the lack of integrated technology solutions. This lack of integration makes communication barriers between healthcare, education systems, and NGOs exist.

User Needs:

- **Streamlined Access to Screening Tools:** Parents and caregivers require a digital, user-friendly centralized platform supporting both mobile and web interfaces to access standardized screening tools (M-CHAT-R) conveniently
- **Automated Scoring and Clear Risk Insights:** Users need a system that automatically calculates screening scores to instantly classify risk levels into Low, Medium, or High Risk, eliminating human calculation errors on complex algorithms
- **Automated Reminders and Analytical Reports:** Users require dynamic notification mechanisms to alert them when re-screening is necessary and actionable visual charts/reports that can be seamlessly downloaded or shared with therapists and doctors.

Challenges:

- **Ineffective Communication Channels:** Without an integrated platform, parents struggle to transfer critical biodata (based on BPPOKU form standards) and screening history back and forth between NGOs and clinics, leading to missed intervention opportunities.
- **Inconsistent Behavior Tracking:** Caregivers face massive difficulties analyzing triggers or recurring mood patterns due to scattered or undocumented emotional records, which significantly undermines the quality of information provided during doctor consultations.

10.1 : Current business process (scenarios, workflow)

Based on the gathered data, the current system (AS-IS) relies heavily on manual processes and traditional methods. We have identified two main users involved in the current workflow: Parents/Caregivers and Clinic Administrative Staff. Each of them goes through a manual process that currently lacks integration and centralization.

When parents notice unusual behavior observations at home, they must track daily moods and emotions manually, capturing unusual emotion and behavior records in a physical notebook or relying on memory. This leads to scattered emotional records over time. To proceed with formal screening, parents must secure a blank physical M-CHAT-R questionnaire from the Clinic paper inventory. The parent then completes the physical forms and attaches the BPPOKU biodata by hand, leaving the clinic with poorly stored physical records.

Once submitted, clinic administrative staff rely on hand-calculated risk level insights derived from the physical forms. During medical consultation and screening, parents often present an inconsistent behavioral history to the doctor, who must review these inconsistent patient records. If diagnostic feedback and validation are confirmed, healthcare providers issue physical referral documents and requests back to the parents and the clinic admin staff. Finally, the admin staff manually files patient data and schedules updates within a clinic physical cabinet, a fragmented process that ultimately subjects parents to disorganized follow-up screening reminders.

Workflow

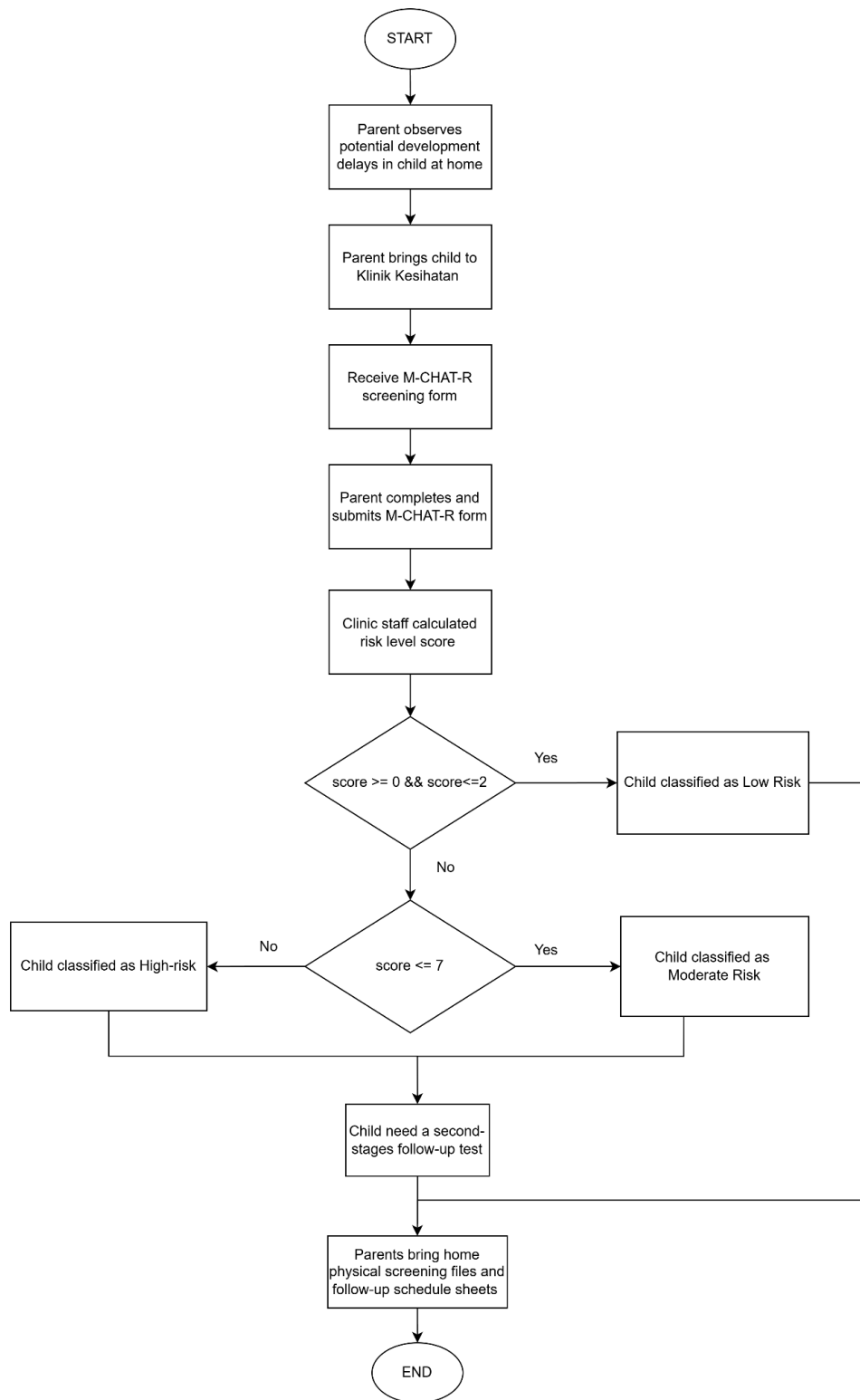


Diagram 1:Workflow

10.2 : Functional Requirement (input,process,output)

10.2.1 : Context Diagram

Process	Input	Output
Current Manual Autism Screening and Behavioral Tracking System	- Unusual behavior observations (Parents) - M-CHAT-R forms (Parents) - Patient identification - Patient file (Clinic Admin Staff)	- Emotional & behavioral records - Patient records - Follow-up screening schedule - Patient screening files

Table 1:Context Diagram

10.2.2 : Diagram 0

Process	Input	Output
Record Daily Mood and Emotions	- Unusual behavior observations (parents)	- Emotional and behavioral records (D1)
Conduct Autism Screening Process	- M-CHAT-R forms (parents) - Patient identification	- Patient screening files (D2) - Patient records
Manage Patient Data and Follow-up Scheduling	- Patient file (Clinic admin staff)	- Follow-up screening schedule (D3)

Table 2:Diagram 0

10.2.3 : Child Diagram

10.2.3.1: Record Daily Mood and Emotions

Process	Input	Output
Observe Behavioral Patterns	- Unusual behavior observations	- Behavioral observation details
Write Informal Logs	- Behavioral Observation details	- Emotional and behavioral records (D1)

Table 3:Record Daily Mood and Emotions

10.2.3.2 : Conduct Autism Screening Process

Process	Input	Output
Fill in Assessment Forms	- M-CHAT-R forms - Patient identification	- M-CHAT-R forms (filled)
M-CHAT-R Analysis & Scoring	- M-CHAT-R forms (filed)	- M-CHAT-R forms result and risk level
Manually Calculate Risk Levels	- M-CHAT-R forms result and risk level	- Patient screening files (D2)

Table 4:Conduct Autism Screening Process

10.2.3.3: Manage Patient Data and Follow-up Scheduling

Process	Input	Output
File and Archive Hardcopies	- Patient file - Patient screening files (D2)	- Archived patient records (D2)
Track Dates and Issue Reminders	- Archived patient records (D2)	- Follow-up screening schedule (D3)

Table 5: Manage Patient Data and Follow-up Scheduling

10.3 Non-functional Requirement (performance and control)

Performance	Control	Requirement Description
Speed	User/event response time	The automated M-CHAT-R scoring engine shall classify risk (Low/Moderate/High) within 2 seconds after the parent submits the final answer.
Speed	Screen refresh time	The Behavior Monitoring mood trend chart (last 30 days) shall load and refresh within 3 seconds.
Reliability	Availability	The system shall be available to parents and NGO admins during normal working hours (Mon-Fri, 8:00 – 20:00) with 99.5% availability (downtime $\leq$ 3 minutes per day).
Reliability	Rate of failure occurrence	To solve the current manual scoring error rate (15–20% from AS-IS analysis), the automated scoring engine shall achieve a failure rate of $\leq 0.5\%$ (99.5% accurate risk classification).
Ease of use	Training time	Parents without technical training shall complete the 20 question M-CHAT-R screening with 0 minutes of formal training required.
Ease of use	Number of help frames	Each major module (Screening, Behavior Monitoring, Profile) shall provide at least 1 help frame or tooltip to guide users.

Robustness	Time to restart after failure	In the event of system failure during screening, the system shall restart and restore the user's previous session within 5 minutes.
Robustness	Percentage of events causing failure	The system shall handle concurrent access by up to 50 users without failure. Percentage of events causing failure shall be $\leq 1\%$.
Portability	Number of target systems	The solution shall operate on 2 target systems: Android mobile devices (minimum API level 26) and standard web browsers (Chrome, Firefox, Edge).
Portability	Percentage of target dependent statements	The codebase shall maintain $\leq 15\%$ platform dependent statements to ensure easy cross platform maintenance.

Table 6: Non-functional Requirement (performance and control)

10.4 Logical DFD Diagram

10.4.1 Logical Context Diagram

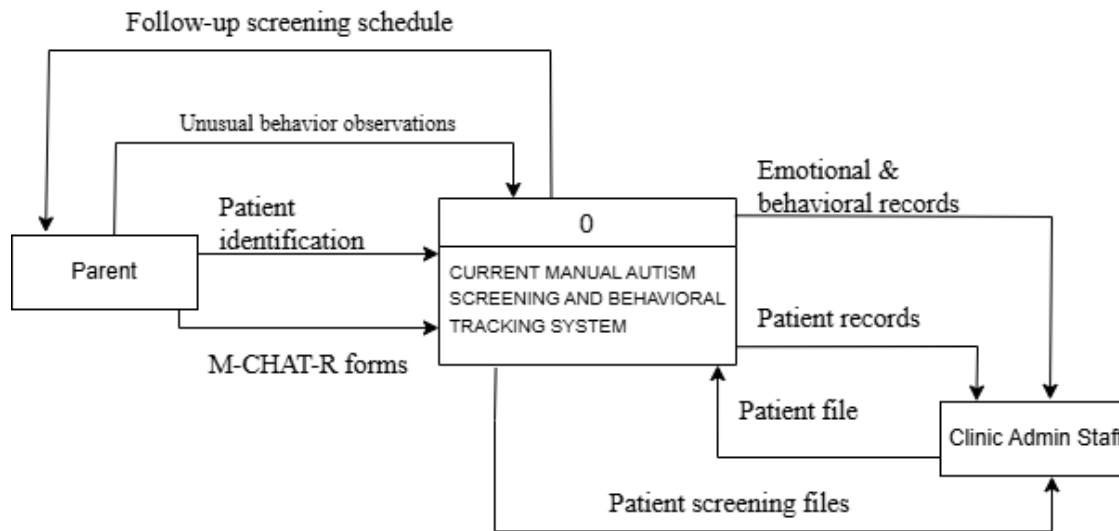


Diagram 2: Logical Context Diagram

This Context Diagram maps the core interactions between external entities and the manual autism screening system. It outlines the foundational flow of patient forms, records, and scheduling info.

10.4.2 Logical Diagram 0

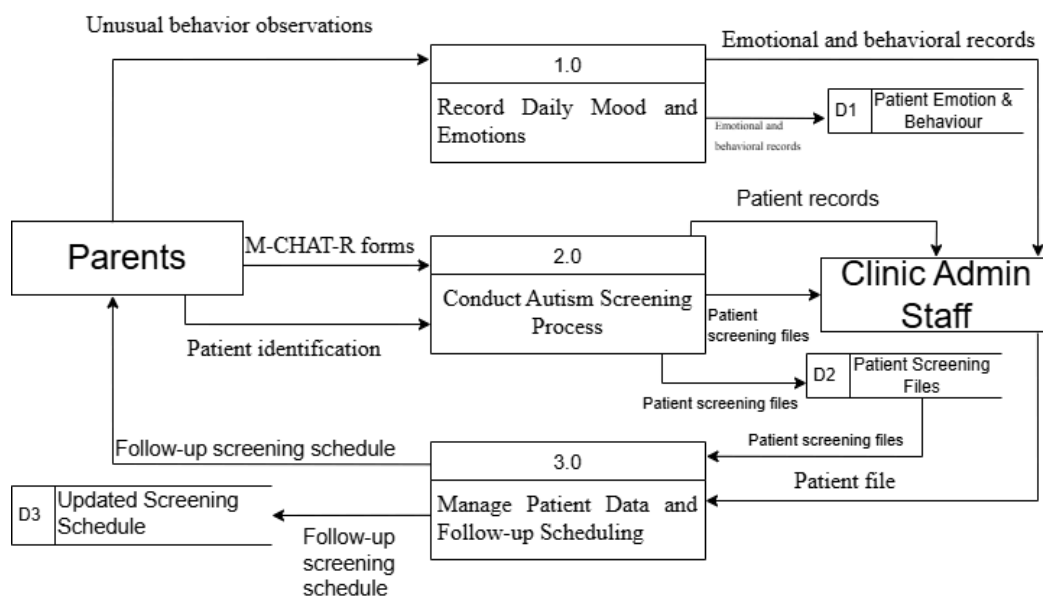


Diagram 3: Logical Diagram 0

This diagram breaks down the main system into three primary processes which is recording daily mood or emotions, conducting the screening process, and managing patient data. It also introduces local data stores where files and screening schedules are kept.

10.4.3 Child Diagram

We have prepared three child diagrams for each process including recording daily mood and emotions, conducting an autism screening process and managing patient data and follow up scheduling.

10.4.3.1 Record Daily Mood and Emotions

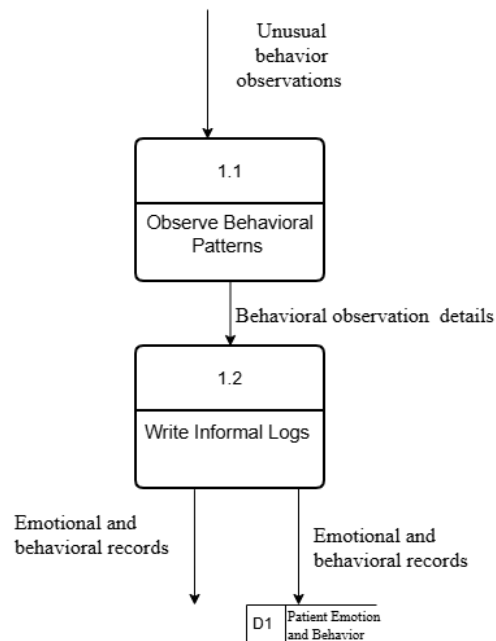


Diagram 4: Record Daily Mood and Emotions

First Child-level diagram detailing how unusual behavior observations are analyzed. It shows the flow from observing behavioral patterns to writing informal logs, which are then saved to the patient emotion and behavior data store.

10.4.3.2 Conduct Autism Screening Process

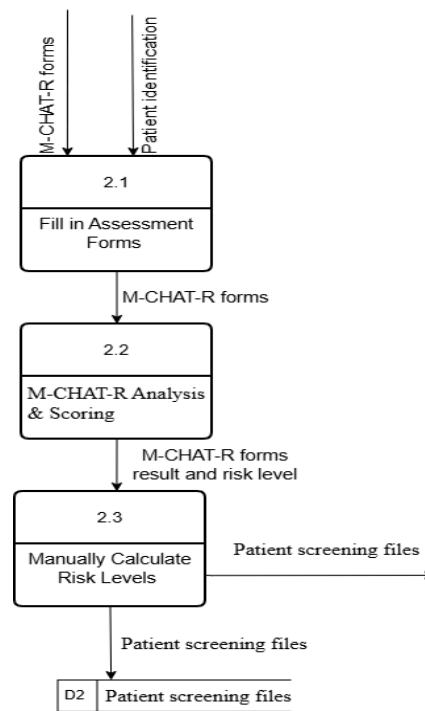


Diagram 5:Conduct Autism Screening Process

Second Child-level diagram expands process 2.0 to show the progression of the screening workflow. It displays how input assessment forms are analyzed, scored, and used to manually calculate risk levels before being filed into the screening data store .

10.4.3.3 Manage Patient Data and Follow-up Scheduling

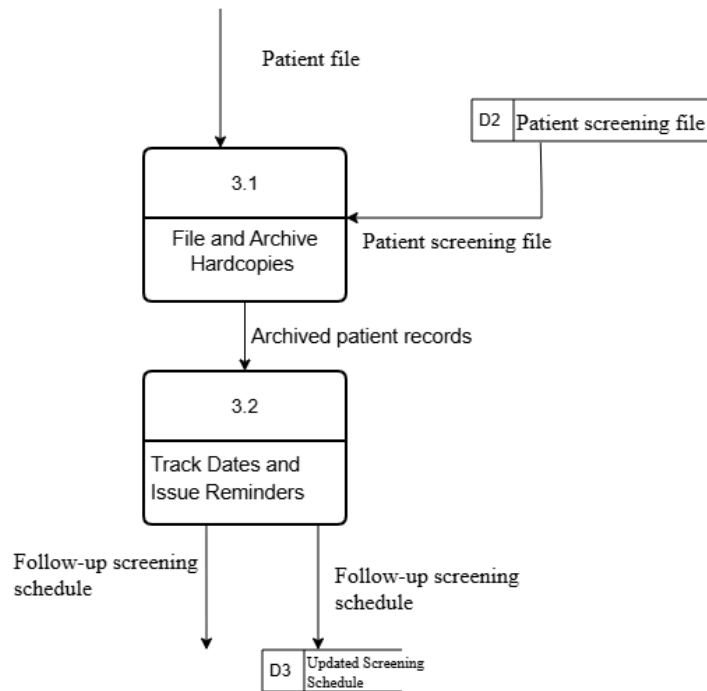


Diagram 6: Manage Patient Data and Follow-up Scheduling

The last Child-level diagram expands process 3.0, illustrating the administrative side of the system. It maps how physical hard copies are archived and how dates are tracked to issue follow-up screening reminders to parents.

11.0 Summary of Requirement Analysis process

The requirement analysis process was systematically conducted to address the gaps identified in the Background Study and Problem Statement. The Background Study revealed that despite the availability of validated tools like the M-CHAT-R, real-world screening is plagued by manual paperwork, disconnected systems between parents and NGOs, and a lack of continuous daily support. The Problem Statement further structured these issues into three operational failures: (1) manual scoring errors and diagnostic delays, (2) data fragmentation with no alignment to BPPOKU regulatory standards, and (3) the absence of automated follow up reminders. By examining the manual AS-IS business scenarios alongside three primary stakeholder groups, Parents/Caregivers, NGO/Clinic Administrative Staff, and Healthcare Providers. The project team successfully captured, categorized, and translated real world pain points into an integrated, cross platform digital lifecycle.

1. Problem Identification & User Need Synthesis

Through targeted information gathering (including a Google Form questionnaire with 16 respondents) and workflow evaluation, the analysis confirmed the following critical system failures:

- Operational Inefficiencies: Manual scoring of physical M-CHAT-R forms led to calculation errors and process delays, causing many families to miss the critical 16 to 30 month early intervention window.
- Data Fragmentation: Initial screening data was isolated and structurally disconnected from official regulatory formats, specifically the BPPOKU Part A (Amendment 2019) standards.
- Lack of Proactive Action: Follow up tracking for moderate or low risk children was highly unstable due to the absence of automated monitoring and reminder systems.

To counter these challenges, user requirements were emphasized around three main pillars, providing streamlined digital access to screening tools, implementing an automated scoring engine to instantly classify risk levels (Low, Moderate, High), and deploying automated reminder mechanisms alongside downloadable analytical reports.

2. Functional Architecture Blueprinting (DFD Transformation)

The captured functional requirements were formally modeled using structured Systems Analysis and Design techniques. This began with a Context Diagram to map core system interactions and external entities. It was then broken down into a Diagram 0 and Child Diagrams across six vital system capabilities:

· **Manage Authentication & Registration:** Ensuring secure, role-based access for parents and administrators.

- **Manage Profile:** Structuring child biodata, birth certificates, and family backgrounds completely around official BPPOKU data schemas to eliminate duplicate data entries.
- **Manage Screening:** Transforming the 20 question M-CHAT-R instrument into an interactive digital format that eliminates human calculation errors through instant automated scoring.
- **Manage Notification & Reminder:** Providing automatic push notifications and email mechanisms to protect developmental timelines and reschedule rescreenings.
- **Manage Reporting & Analytics:** Equipping NGO administrators with centralized dashboards to track community trends and monitor formal diagnostic referrals.
- **Manage Behavior Monitoring (Add-On Module):** Transitioning manual, scattered logging into an automated platform where caregivers can record daily moods, sleep patterns, and routines to render long-term visual trend charts.

In summary, the requirement analysis process successfully bridged the gaps identified in the Background Study and Problem Statement by defining a clear, actionable blueprint for a digital system that is technically feasible, operationally practical, and economically viable.

12.0 System Analysis and Specification

12.1 Logical DFD To-Be System

12.1.1 Context Diagram

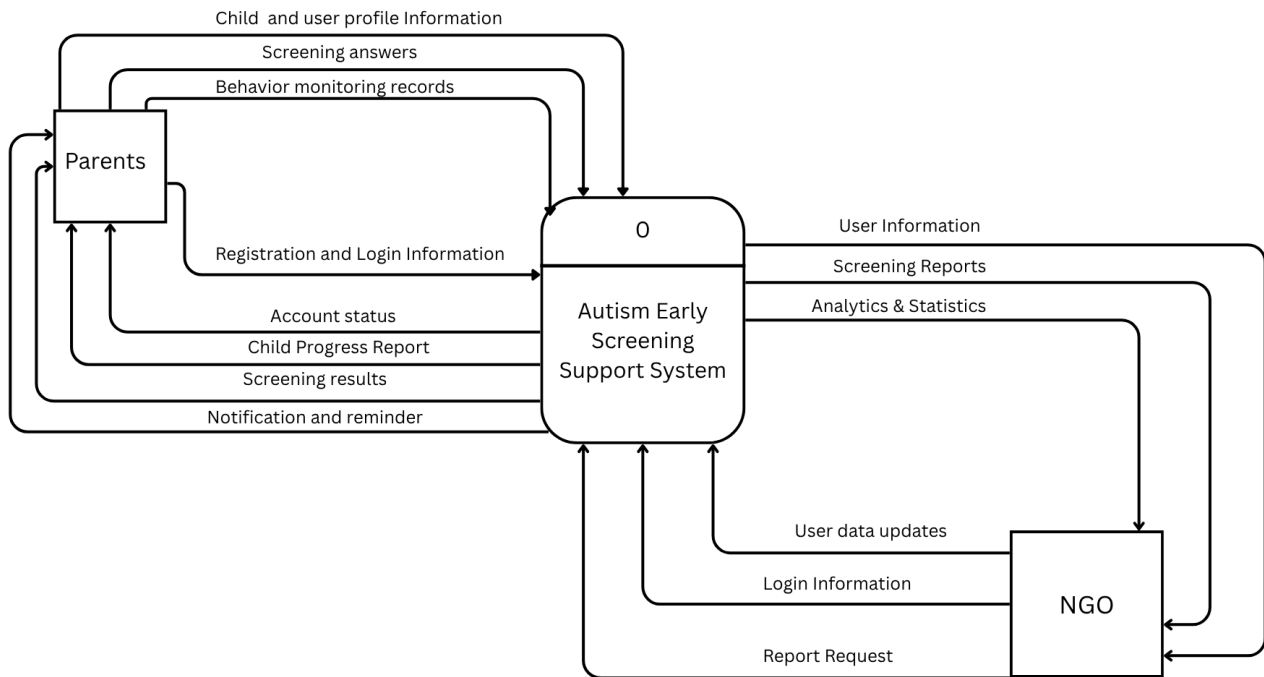


Diagram 7:Context Diagram To-Be

This Context Diagram for the Autism Early Screening Support System acts as an intermediary between two external entities.

12.1.2 Diagram 0

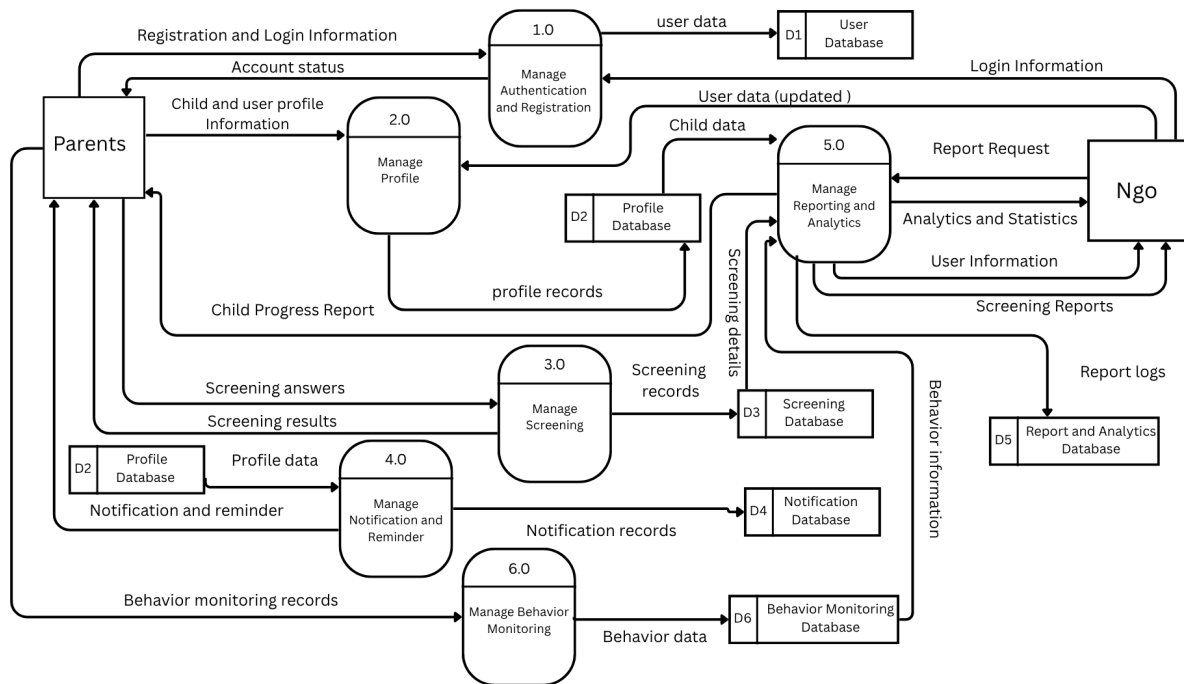


Diagram 8:Diagram 0 To-Be

This diagram illustrates the data flow that breaks open the central system to show how information is processed across six subprocesses and stored in six data stores.

12.1.3 Child Diagram

12.1.3.1 Manage Authentication and Registration

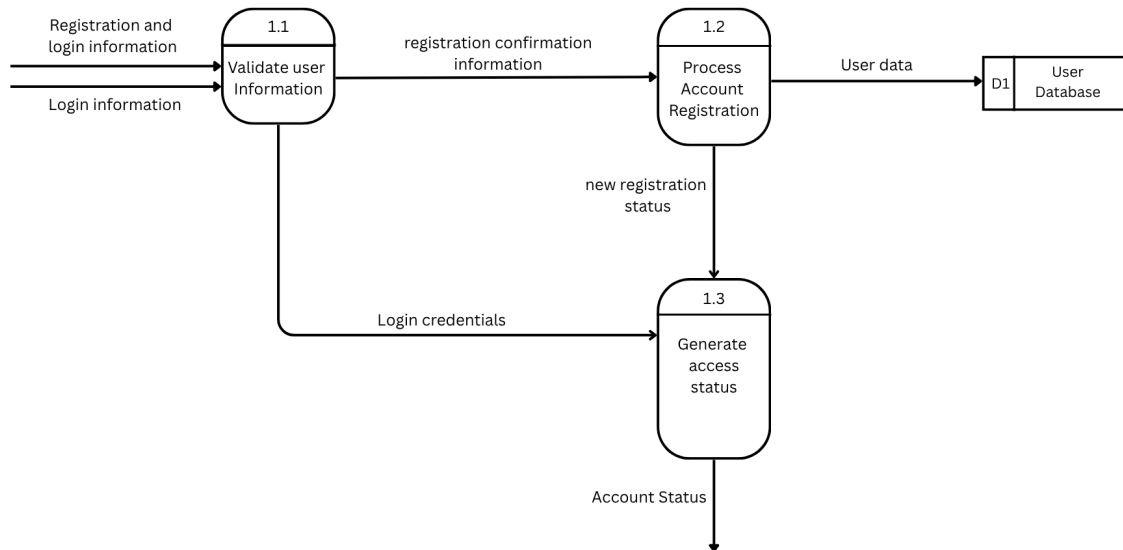


Diagram 9: Manage Authentication and Registration

This diagram shows the flow of the user registration and login process by receiving and checking the validity of the information sent. After verification, the successfully registered user data will be officially stored in the database. The credential information is then used to generate the account access status before being sent out as the final output to the user.

12.1.3.2 Manage Profile

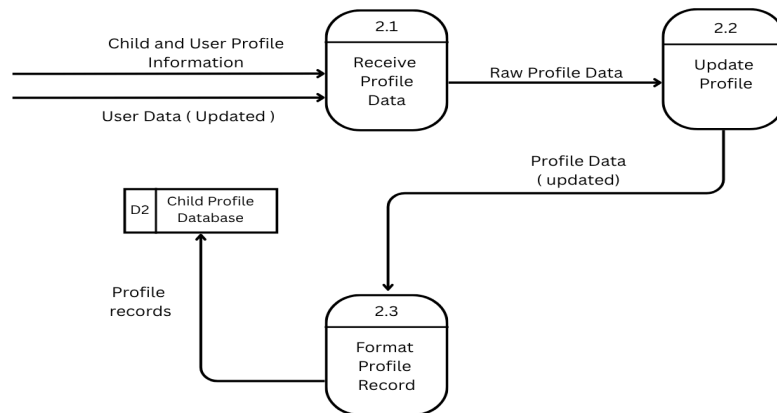


Diagram 10: Manage Profile

The diagram shows the profile management process by receiving child and user profile information, including updated user data. This raw information received is then processed to update existing profile data in the system. Finally, the updated profile data is formatted into a structured record before being securely stored in the profile database.

12.1.3.3 Manage Screening

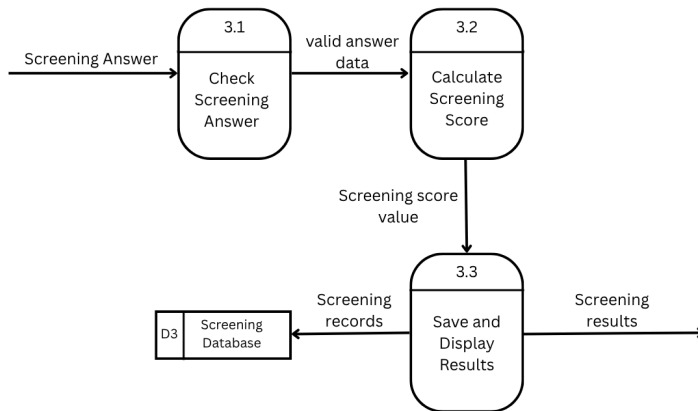


Diagram 11: Manage Screening

This diagram explains the management of the screening test by receiving and checking the validity of the submitted screening answers. The validated answer data is then processed to calculate the total screening score value obtained. Finally, the results will be stored in the screening database and named as a screening result output display to the user.

12.1.3.4 Manage Notifications and Reminder

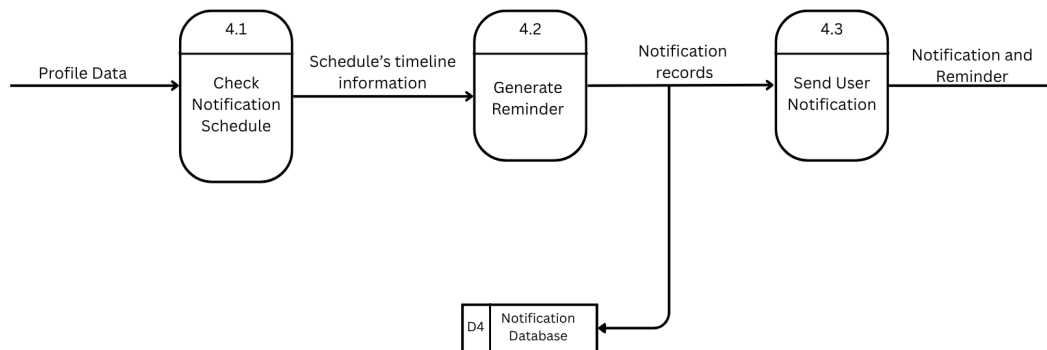


Diagram 12: Manage Notifications and Reminder

This diagram shows the operation of notification and reminder management by checking the notification schedule based on the incoming profile data. The obtained schedule timeline information is then used to generate appropriate reminders for the user. Finally, the notification record will be stored in the notification database and sent out as notification and reminder output to the user.

12.1.3.5 Manage Reporting and Analytics

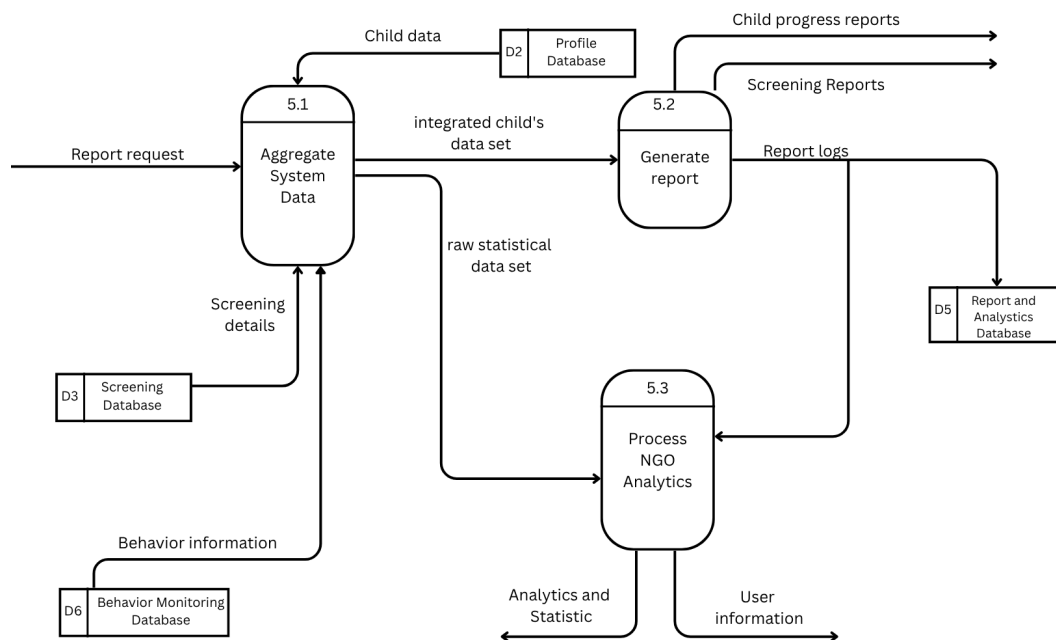


Diagram 13: Manage Reporting and Analytics

This diagram illustrates reporting and analysis management by collecting and integrating various system data from profiling, screening, and behavioral monitoring databases based on incoming report requests. The integrated data is then processed to generate child progress reports and screening reports, where the resulting report logs are stored in the reporting and analysis database. Finally, the report logs along with this raw statistical data set are analyzed to produce analytical outputs, statistics, and user information for the needs of relevant parties.

12.1.3.6 Manage Behavior Monitoring

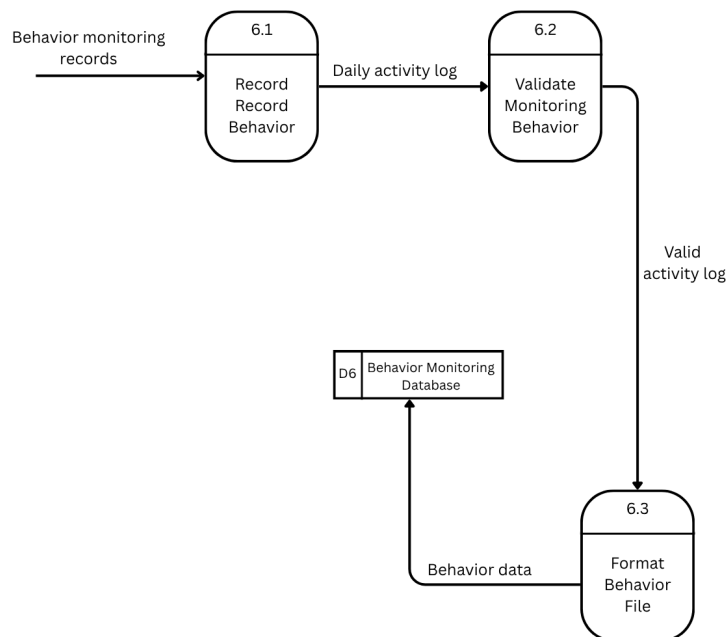


Diagram 14: Manage Behavior Monitoring

This diagram shows behavior monitoring by receiving and recording daily activity log data from incoming behavior monitoring records. The daily activity log is then filtered and validated to ensure that the information collected is valid. Finally, the validated activity log is formatted into a structured behavior data file before being officially stored in the behavior monitoring database.

12.2 Process Specification

12.2.1 Manage Authentication and Registration

Process Specification Items	Description
Process Number	1.0
Process Name	Manage Authentication and Registration
Process Description	Manage the entire process of verifying the identity of old users (login) as well as registering information for new users into the system, subsequently issuing valid access status

Input Data Flow	1. Registration and login information (from Parents) 2. Login information (from Ngo)
Output Data Flow	1. Account status (to Parents) 2. User data (to D1: User Database)
Type of Process	Online
Name of Subprogram	- Validate user information - Process Account Registration - Generate Access Status
Process Logic Description	-
Logic Methode Reference	Structured English
Unresolve Issues	None. (The system is assumed to always be able to connect to D1: User Database without network interruption).

Table 7: Manage Authentication and Registration

Structured English

START

IF user requests registration THEN

 READ Registration and login information

 VALIDATE registration details format

 IF details are valid THEN

 GENERATE registration confirmation information

 CREATE new user account record

 WRITE User data TO D1: User Database

 SET new registration status to 'Success'

 GENERATE Account status as 'Registration Successful / Active'

 ELSE

 GENERATE Account status as 'Registration Failed'

ENDIF

ELSE IF user requests login THEN

 READ Login information

 VALIDATE login credentials format

 IF format is valid THEN

 VERIFY credentials against D1: User Database

 IF credentials match record THEN

 GENERATE Account status as 'Login Successful'

 ELSE

 GENERATE Account status as 'Invalid Credentials'

 ENDIF

 ELSE

 GENERATE Account status as 'Invalid Format'

 ENDIF

ENDIF

OUTPUT Account status to user

END

12.2.2 Manage Profile

Process Specification Items	Description
Process Number	2.0
Process Name	Manage Profile
Process Description	Manages the receipt of new profile information or updates to existing profiles from parents or NGOs, processes the updates, and converts them to the appropriate record format before storing them into the profile database.
Input Data Flow	1. Child and User Profile Information (from Parents) 2. User Data (Updated) (from Ngo)
Output Data Flow	1. Profile records (to D2: Child Profile Database)
Type of Process	Online
Name of Subprogram	- Receive Profile Data

	- Update Profile - Format Profile Record
Process Logic Description	-
Logic Methode Reference	Structure English
Unresolve Issues	Confirm whether old profile data needs to be saved (history log data) before being overwritten with new updated data.

Table 8: Manage Profile

Structured English

START

IF Parents submit Child and User Profile Information THEN

 READ Child and User Profile Information

 GENERATE Raw Profile Data

ELSE IF Ngo submits User Data (Updated) THEN

 READ User Data (Updated)

 GENERATE Raw Profile Data

ENDIF

IF Raw Profile Data is available THEN

 PROCESS and apply updates to the profile fields

 VALIDATE data integrity rules

 GENERATE Profile Data (updated)

 COMPLIANCE check and format data into database schema records

 GENERATE Profile records

 WRITE Profile records TO D2: Child Profile Database

ENDIF

END

12.2.3 Manage Screening

Process Specification Items	Description
Process Number	3.0

Process Name	Manage Screening
Process Description	Manages the receipt of screening answers from users, checks the validity of answers, calculates scores based on screening criteria, as well as keeps records and displays screening results to users
Input Data Flow	1. Screening Answer (from Parents)
Output Data Flow	1. Screening results (to Parents) 2. Screening records (to D3: Screening Database)
Type of Process	Online
Name of Subprogram	- Check Screening Answer - Calculate Screening Score - Save and Display Results
Process Logic Description	-
Logic Methode Reference	Strutured English
Unresolve Issues	none

Table 9: Manage Screening

Structured English

START

READ Screening Answer

VALIDATE Screening Answer submission

IF Screening Answer is complete and valid THEN

 GENERATE valid answer data

 COMPUTE Screening score value based on assessment formula

 GENERATE Screening score value

 PREPARE database record from score value

 GENERATE Screening records

 WRITE Screening records TO D3: Screening Database

 DETERMINE assessment outcome category

 GENERATE Screening results

 OUTPUT Screening results to user

ELSE

OUTPUT error message "Incomplete assessment submission"
ENDIF

END

12.3.4 Manage Notifications and Reminder

Process Specification Items	Description
Process Number	4.0
Process Name	Manage Notification and Reminder
Process Description	Manages notification schedule revisions based on user profile data, generates reminder text or content when the timeline arrives, stores notification log records into the database, and sends the final reminder to the user.
Input Data Flow	1. Profile Data (from D2: Profile Database / Parents)
Output Data Flow	1. Notification and Reminder (to Parents) 2. Notification records (to D4: Notification Database)
Type of Process	Online (Automated)
Name of Subprogram	- Check Notification Schedule - Generate Reminder - Send User Notification
Process Logic Description	-
Logic Methode Reference	Structured English
Unresolve Issues	Determine the delivery method (e.g., whether via mobile app notification, SMS, or email) if the user is not logged into the platform.

Table 10: Manage Notifications and Reminder

Structure English

START

READ Profile Data

PROCESS Check Notification Schedule

IF schedule timeline matches alert condition THEN

 GENERATE Schedule's timeline information

CONSTRUCT reminder message template
GENERATE Notification records

WRITE Notification records TO D4: Notification Database

DISPATCH alert to communication gateway
GENERATE Notification and Reminder
OUTPUT Notification and Reminder to Parents
ENDIF

END

12.3.5 Manage Reporting and Analytics

Process Specification Items	Description
Process Number	5.0
Process Name	Manage Reporting and Analytics
Process Description	Collect and combine data from various databases (profiles, screenings, and behaviors) based on report requests, generate child development reports and screening reports, record report logs, and process statistical analysis of user information specifically for NGOs.
Input Data Flow	1. Report request (from Ngo) 2. Child data (from D2: Profile Database) 3. Screening details (from D3: Screening Database) 4. Behavior information (from D6: Behavior Monitoring Database)
Output Data Flow	1. Child progress reports (to Parents) 2. Screening Reports (to Ngo) 3. Analytics and Statistic (to Ngo) 4. User information (to Ngo) 5. Report logs (to D5: Report and Analytics Database)

Type of Process	Online
Name of Subprogram	- Aggregate System Data - Generate Report - Process NGO Analytics
Process Logic Description	-
Logic Method Reference	Structured English
Unresolve Issues	Ensure privacy controls and data protection clauses are complied with before profile data (User information) is sent in detail to the NGO.

Table 11: Manage Reporting and Analytics

Structured English

START

IF Report request is received THEN

 READ Child data FROM D2: Profile Database

 READ Screening details FROM D3: Screening Database

 READ Behavior information FROM D6: Behavior Monitoring Database

 AGGREGATE data into integrated system structures

 GENERATE integrated child's data set

 GENERATE raw statistical data set

ENDIF

IF integrated child's data set is available THEN

 COMPILE and structure data into visual documents

 GENERATE Child progress reports

 GENERATE Screening Reports

 GENERATE Report logs

 WRITE Report logs TO D5: Report and Analytics Database

ENDIF

IF raw statistical data set AND Report logs are available THEN

 PROCESS and evaluate analytical correlations

 GENERATE Analytics and Statistic

 GENERATE User information

ENDIF

END

12.2.6 Manage Behavior Monitoring

Process Specification Items	Description
Process Number	6.0
Process Name	Manage Behavior Monitoring
Process Description	Receives child behavior monitoring records from parents, filters and verifies the daily activity logs, and converts them into a standard behavior file format before storing them into the database.
Input Data Flow	1. Behavior monitoring records (from Parents)
Output Data Flow	1. Behavior data (to D6: Behavior Monitoring Database)
Type of Process	Online
Name of Subprogram	- Record Record Behavior - Validate Monitoring Behavior - Format Behavior File
Process Logic Description	-
Logic Methode Reference	Structured English
Unresolve Issues	Specify a daily upload limit for behavior logs to avoid data redundancy in the database.

Table 12: Manage Behavior Monitoring

Structured English

START

READ Behavior monitoring records
 PROCESS Record Record Behavior
 GENERATE Daily activity log

READ Daily activity log
 VALIDATE monitoring data attributes

IF Daily activity log is valid THEN
 GENERATE Valid activity log

CONVERT and structure activity log into standardized schema
 GENERATE Behavior data

WRITE Behavior data TO D6: Behavior Monitoring Database

```

ELSE
    REJECT activity log submission
ENDIF

END

```

13.0 Physical System Design

13.1 Physical DFD To-Be System

13.1.1 Physical DFD TO-BE Diagram 0

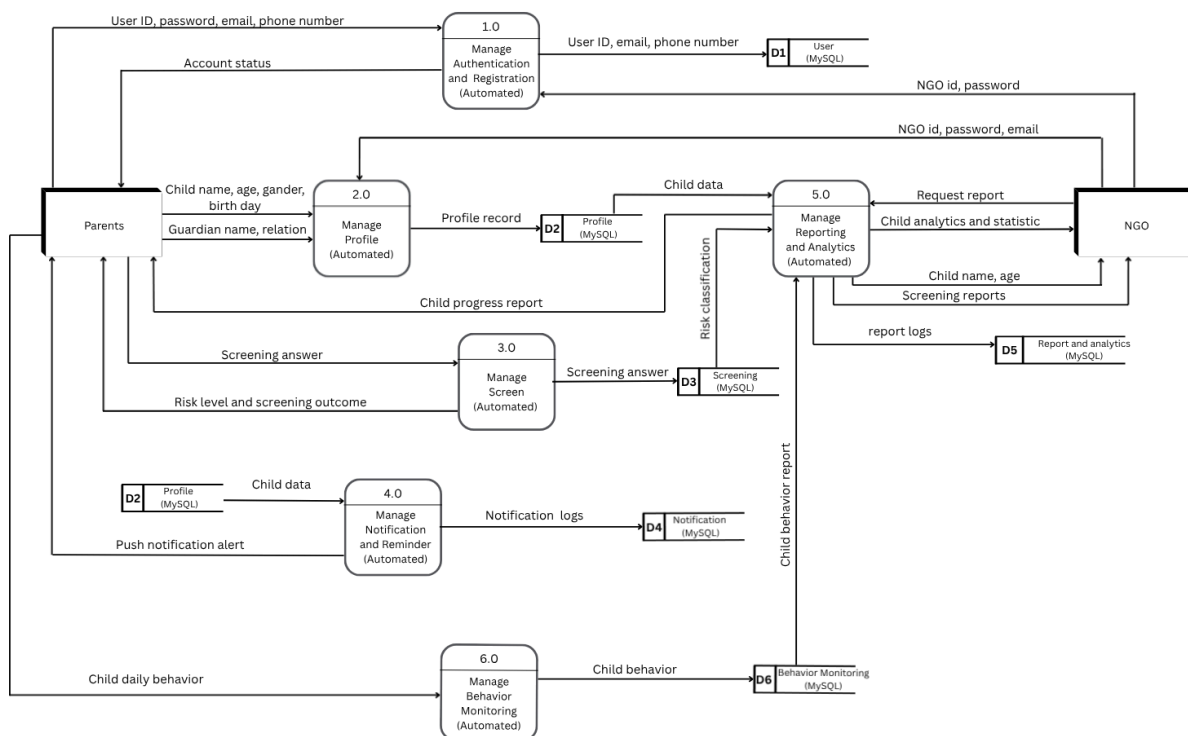


Diagram 14:Physical DFD TO-BE Diagram 0

The diagram illustrates a Physical Data Flow Diagram that shows how they are implemented technically. It explicitly states that all core processes are completely automated and identifies the storage technology as a MySQL relation database across all data stores.

13.1.2 Physical DFD TO-BE Child Diagram

13.1.2 .1 Manage Authentication and Registration

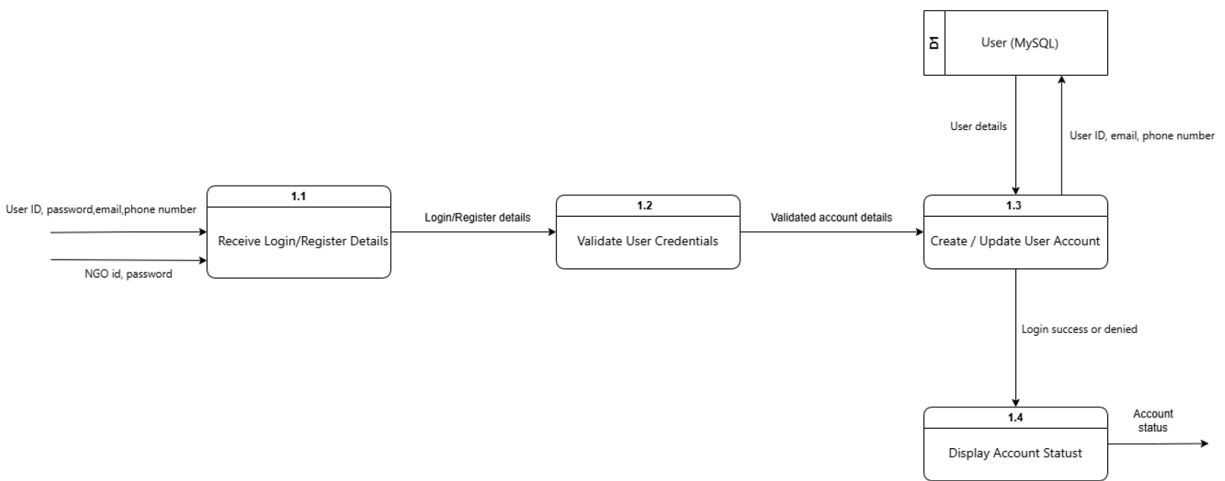


Diagram 15: Manage Authentication and Registration

Diagram maps how users log in or register for an account. It takes raw credentials (like passwords, emails, and IDs), double-checks that they are valid, creates or updates the record in the User database, and returns a simple "success" or "denied" status back to the user's screen.

13.1.2 .2 Manage Profile

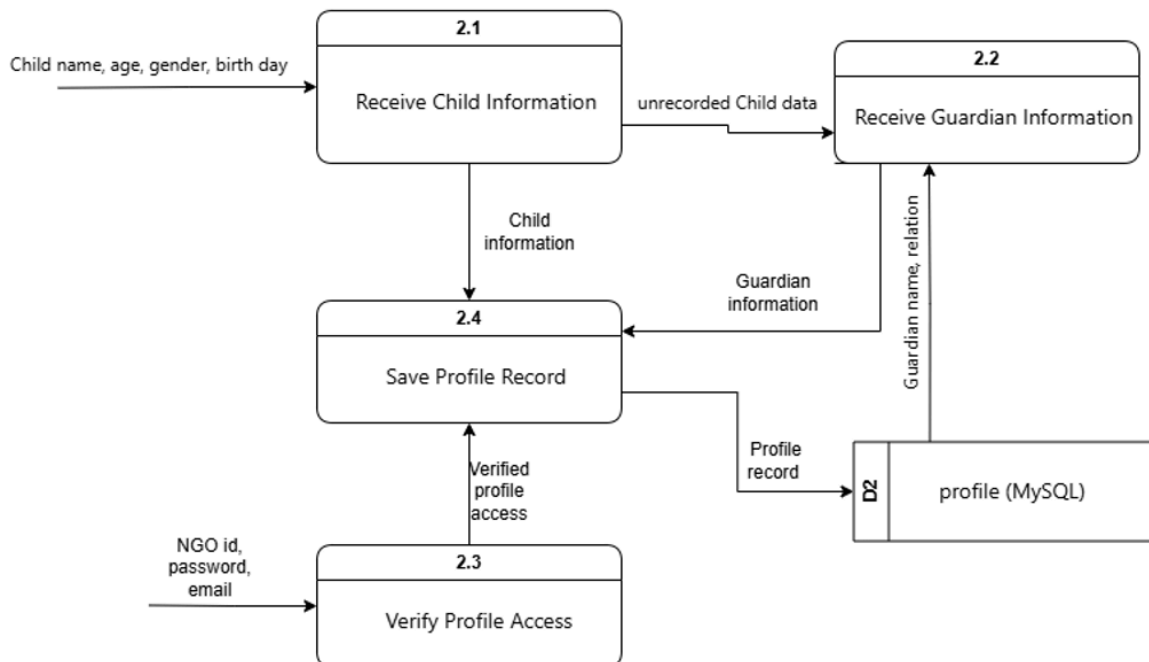


Diagram 16: Manage Profile

Diagram manages the setup of a child's profile in the system. It collects a child's basic demographics, links them to their parent or guardian's information, ensures an authorized NGO user is the one handling the profile, and saves the completed family profile safely into the database.

13.1.2 .3 Manage Screening

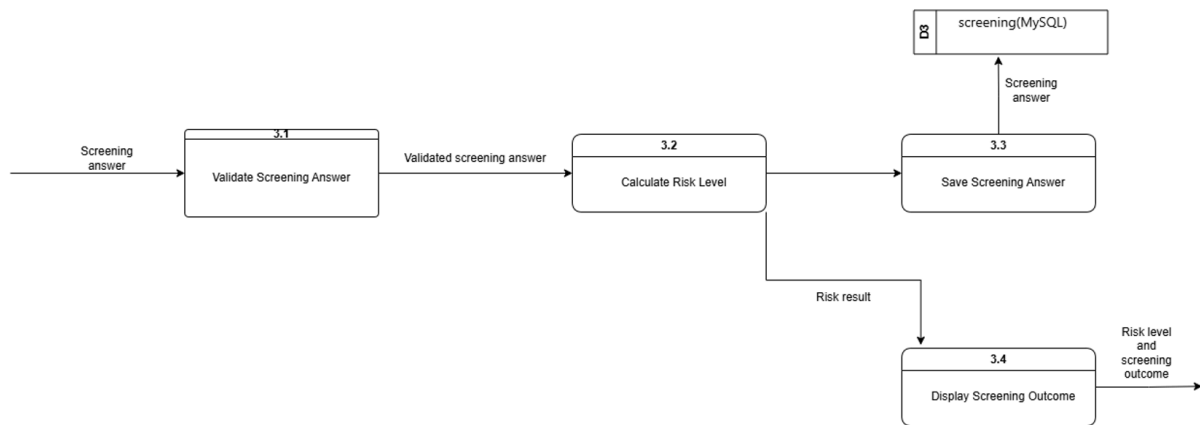


Diagram 17: Manage Screening

Diagram processes the developmental or behavioral screening tests taken for a child. It captures the raw answers given during a test, securely validates them, calculates an automated risk level (e.g., low, medium, high), saves the record for future tracking, and displays the risk result instantly back to the end-use

13.1.2 .4 Manage Notifications and Reminder

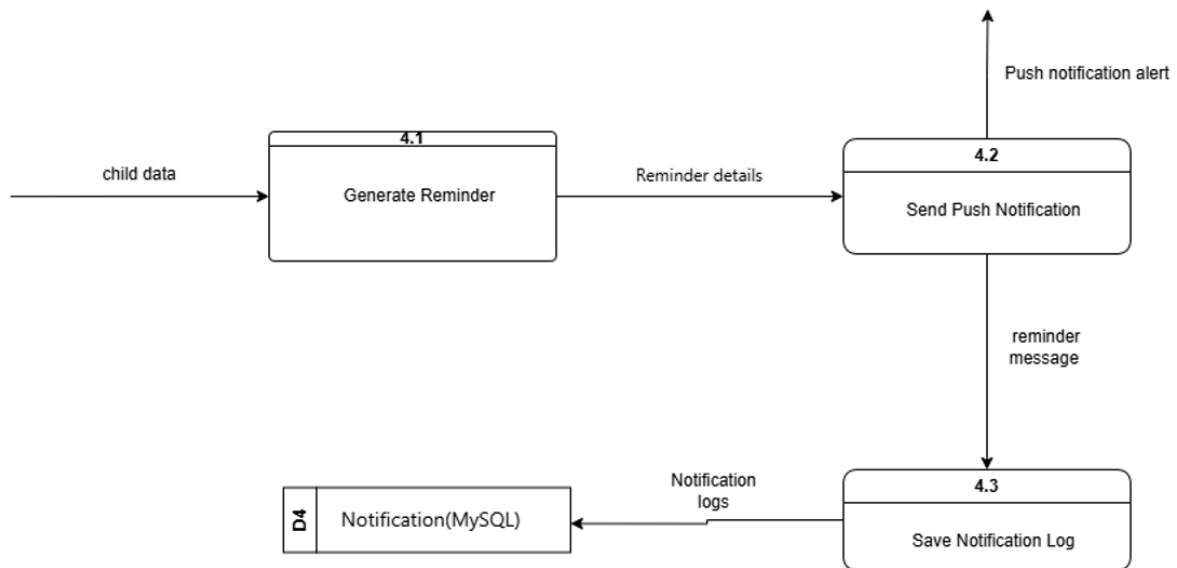


Diagram 18: Manage Notifications and Reminder

Diagram handles the background alerting system. It scans a child's data to check if any critical timelines or milestones are coming up, automatically generates a friendly reminder message, fires it off as an instant push notification alert to a device, and logs the sent message for audit records.

13.1.2 .5 Manage Reporting and Analytics

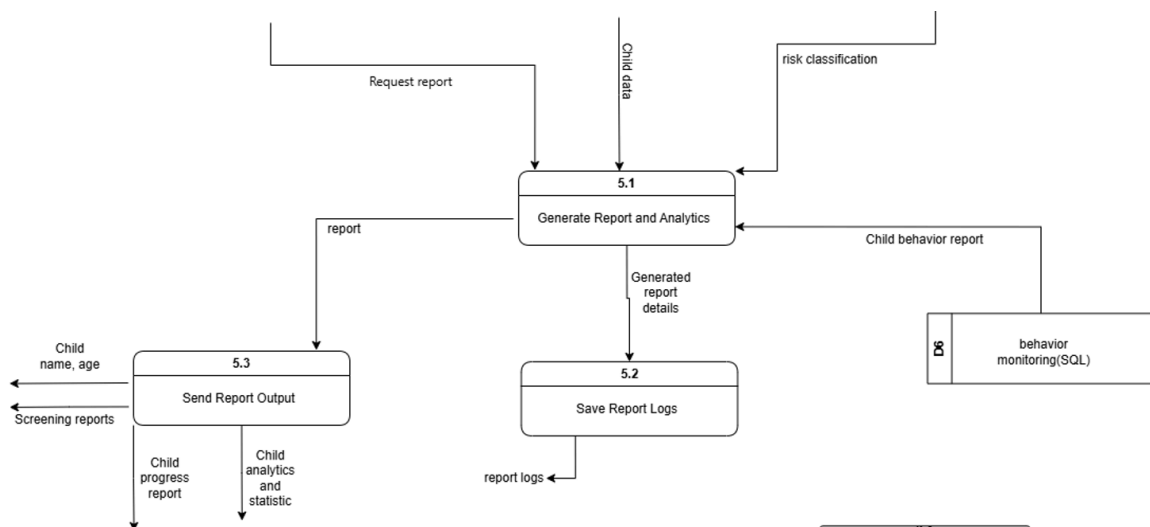


Diagram 19: Manage Reporting and Analytics

Diagram acts as the data compiler for the system. It grabs a mix of different stored records such as behavioral metrics, screening risk historical data, and direct report requests to build readable summaries like progress charts and statistics, saving a record log while sending out the finalized reports.

13.1.2 .6 Manage Behavior Monitoring

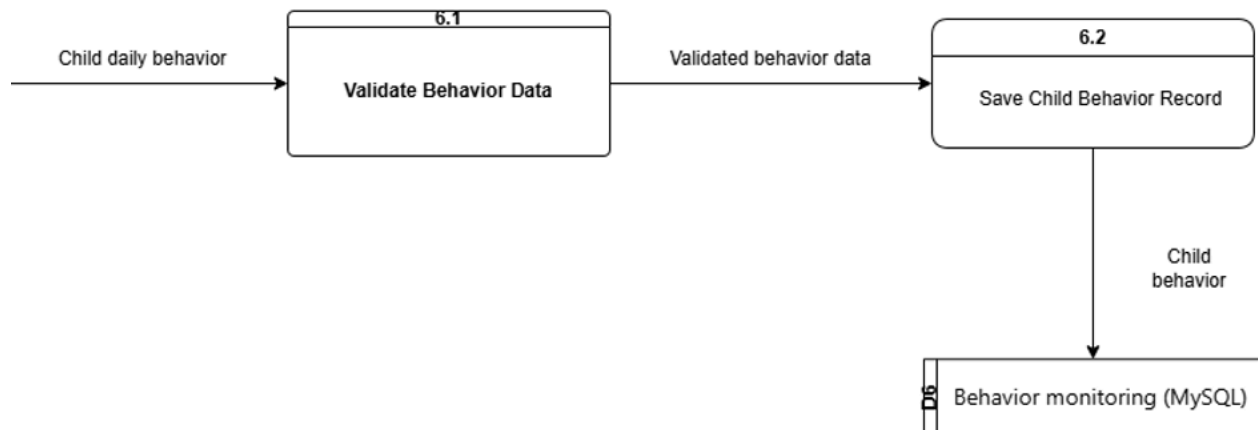


Diagram 20:Manage Behavior Monitoring

Diagram provides a fast tracking mechanism for regular data entry. Whenever a parent or worker observes a child's day to day actions, they submit the behavior data here; the system cleans/checks the entry and immediately updates the ongoing behavior tracking database.

13.1.3 Physical DFD TO-BE Partitioning

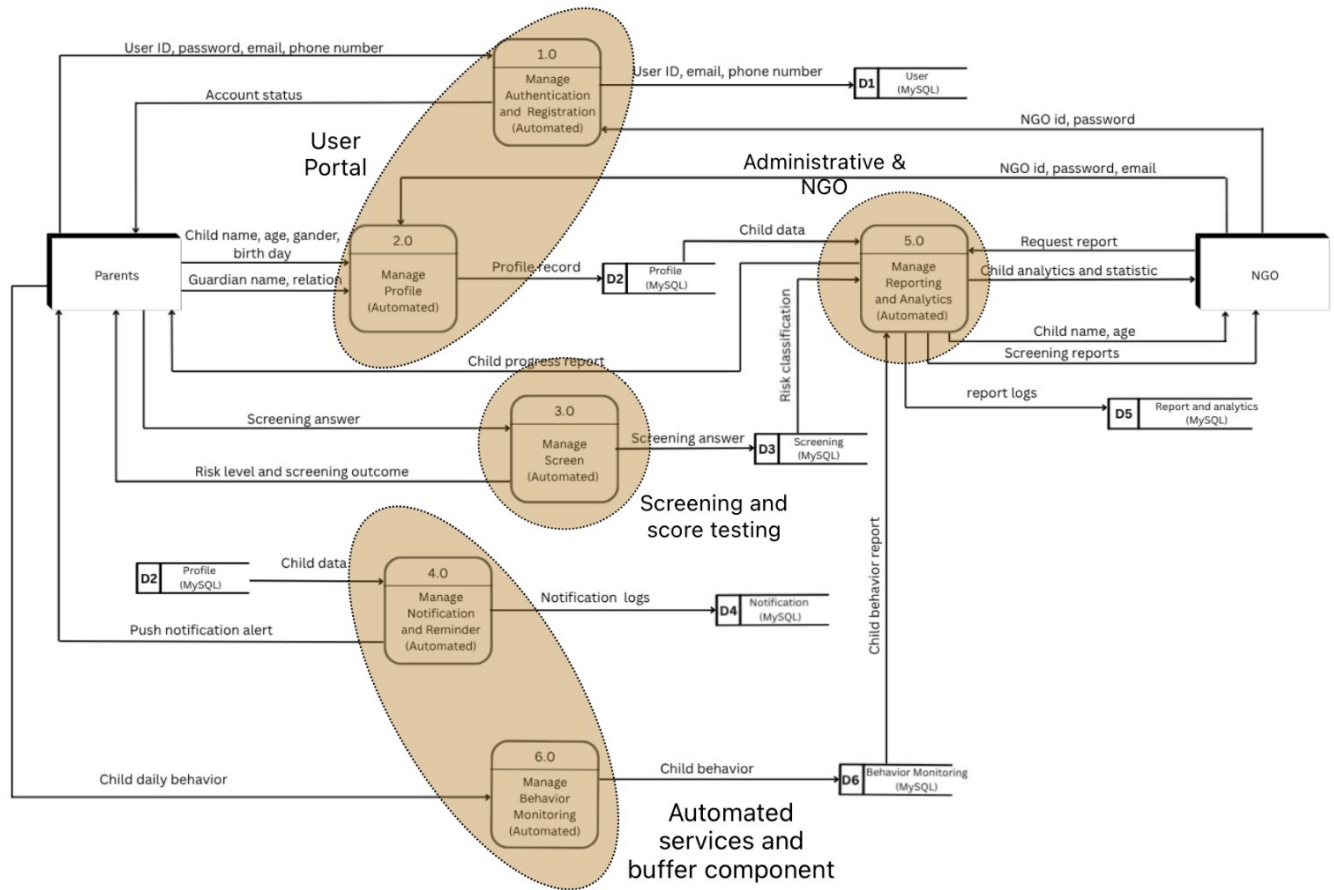


Diagram 21:Physical DFD TO-BE Partitioning

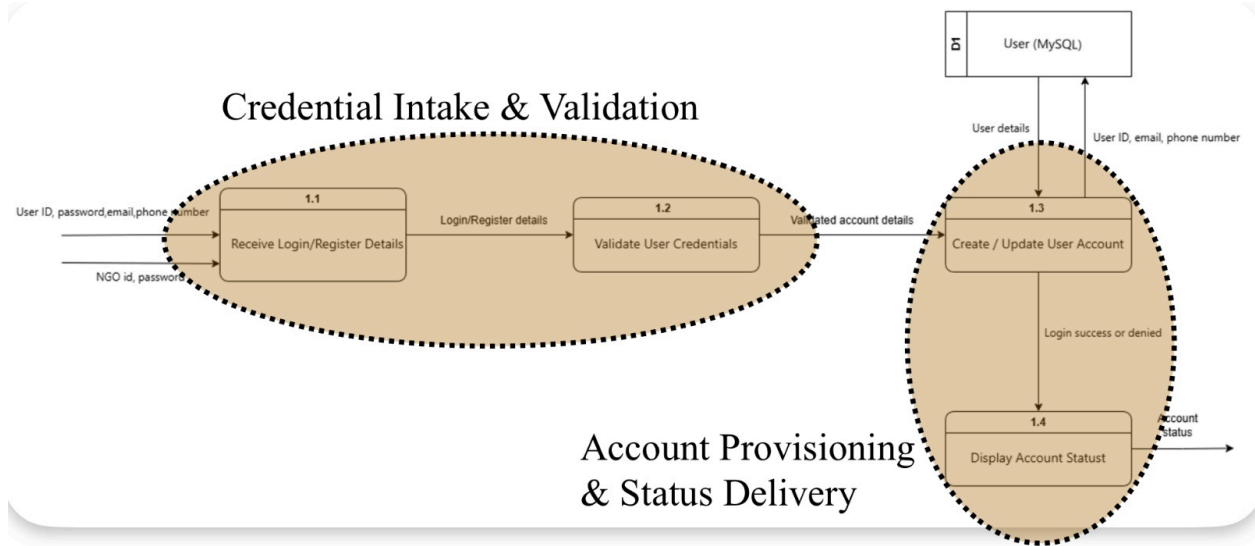


Diagram 22: Manage Authentication and Registration

This partition acts as the security perimeter. It ingests incoming login or registration streams from two distinct roles: regular Users (User ID, password, email, phone number) and NGO personnel (NGO ID, password). It securely processes these credentials through a validation routine before allowing access further inside the system application logic. This partition controls state mutations and feedback. It interfaces directly with the D1 user database to execute structural writes (account creations or updates). It evaluates the database transaction state and translates the query outcomes into an immediate user-facing interface element to let the actor know if the login succeeded or failed.

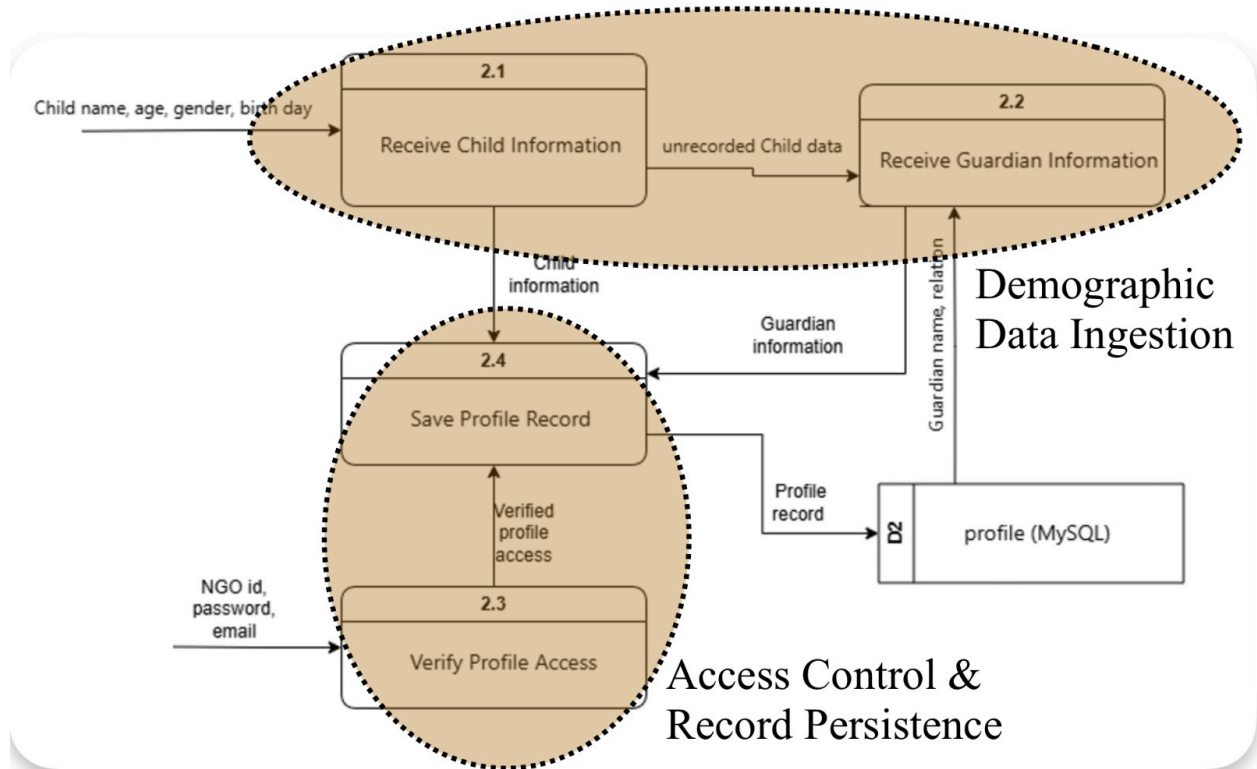


Diagram 23: Manage Profile

This partition handles unrecorded demographic entries. When new child metadata (name, age, gender, birth date) is detected as missing from the system, it automatically spawns a secondary data gathering routine to intercept complementary parent/guardian metadata (guardian name, relationship), binding them together logically as a single unverified family record unit. This partition regulates database writes based on administrative access control. It absorbs NGO authorization signatures (NGO id, password, email) to confirm profile access. Once verified, it merges the staged child and guardian context blocks and commits a unified, validated profile record into the D2 Profile database.

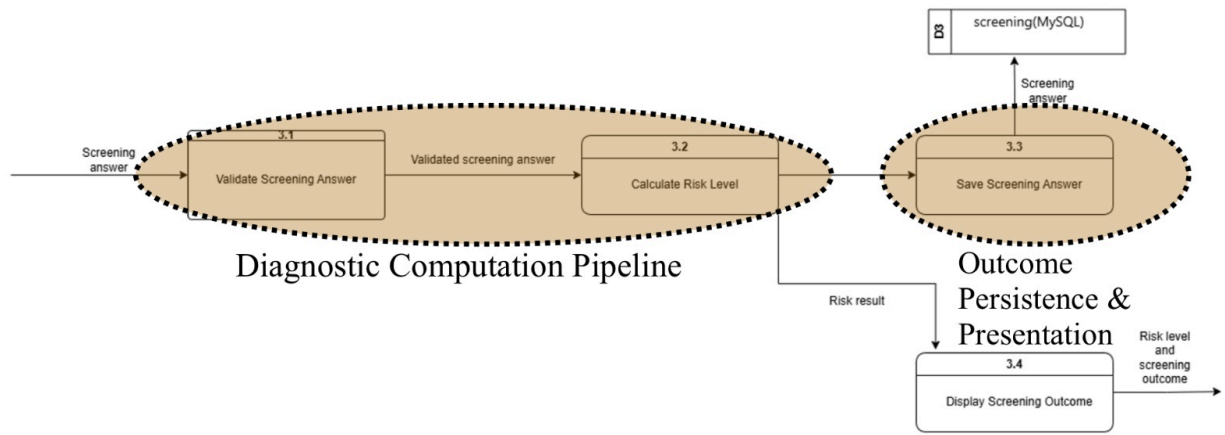


Diagram 24: Manage Screening

This partition represents the primary analytical pipeline. It intercepts screening answers submitted by users, sanitizes the inputs for clinical processing safety, and passes them to a localized rule engine that calculates the behavioral/developmental risk classifications based on the user's answers. This partition splits the calculation results into cold-storage and client presentation vectors. It writes raw screening assessments permanently into the D3 Screening database for historical record-keeping, while simultaneously packaging the processed risk evaluation and piping it directly to the dashboard display UI.

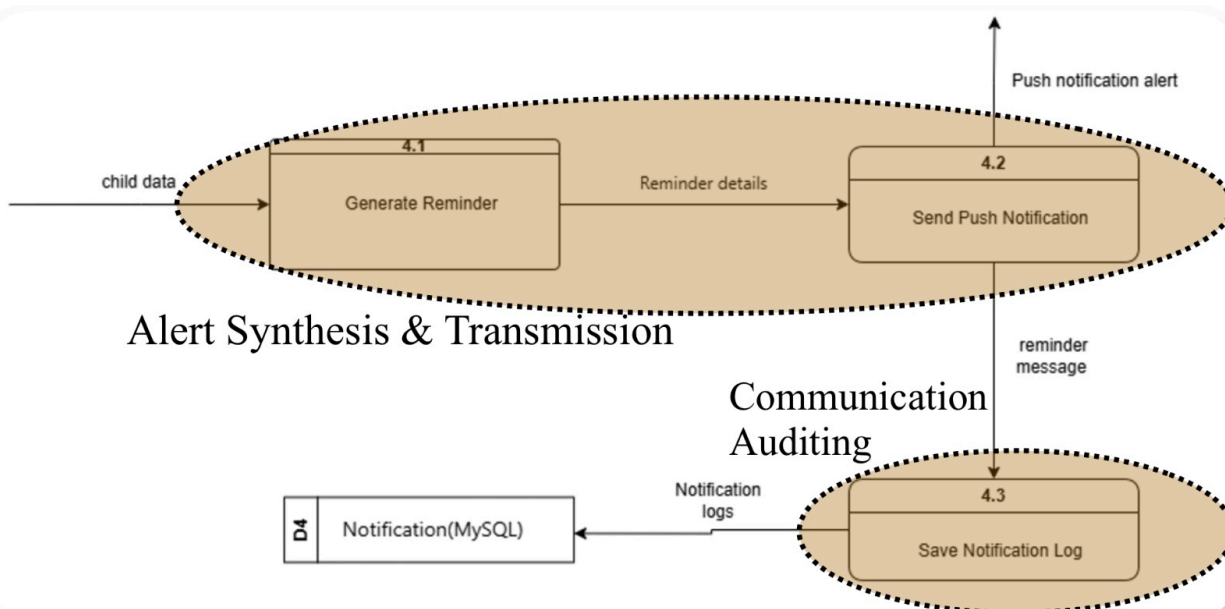


Diagram 25: Manage Notifications and Reminder

A dynamic message broker. It reads existing child transactional logs to evaluate automated timeline conditions. If a criterion matches, it synthesizes an explicit reminder payload (message metadata) and instantly interfaces with system device bridges to fire an outbound push

notification alert. An asynchronous auditing module. It listens exclusively for downstream reminders triggered by the dispatch engine and registers structural transmission logs into the D4 Notification database to track exactly what communications reached the user.

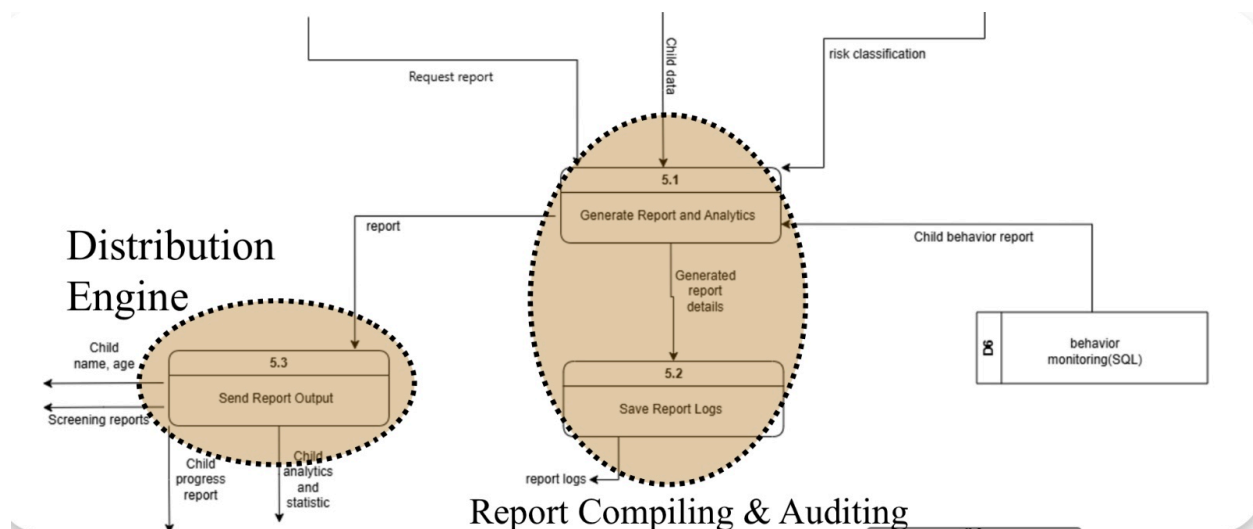


Diagram 26: Manage Reporting and Analytics

A business intelligence processing node. It consumes global asynchronous data flows including explicit user report requests, child demographic records, risk classifications, and behavior histories pulled from the D6 Behavior Monitoring database. It processes this multi-source data into unified system reports while generating operational tracking logs. A downstream distribution node. It receives ready report blocks and branches them into targeted user-facing channels, compiling specific datasets for individual views (Child name/age, historical screening trends, progress tracks, and aggregate child statistics).

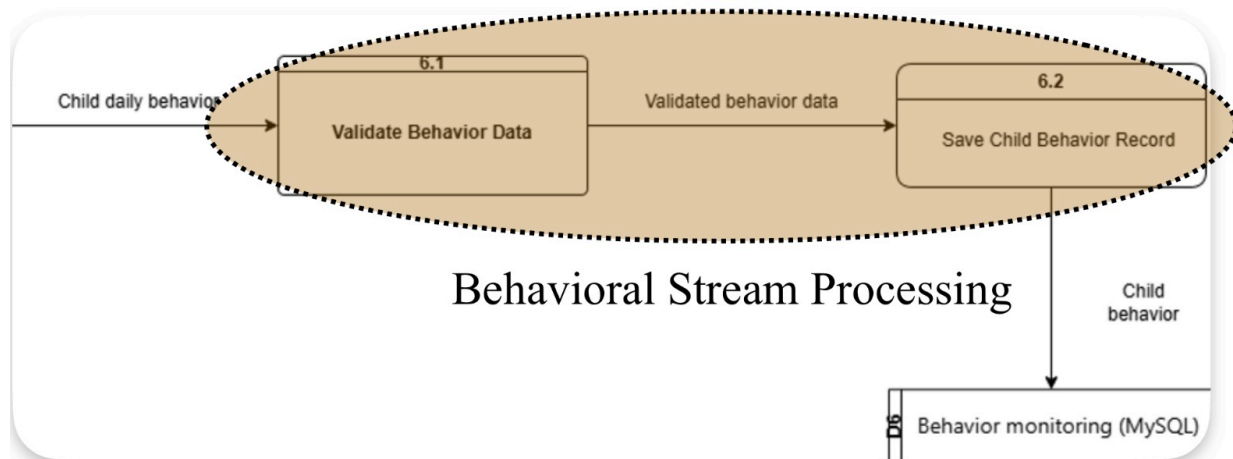


Diagram 27: Manage Behavior Monitoring

A high-throughput, transaction-focused tracking pipeline. It receives constant, real-time telemetry inputs describing a child's day-to-day behavioral activities, runs validation checks to ensure format and range safety, and immediately appends the valid behavioral record into the D6 Behavior Monitoring database.

13.1.4 Physical DFD TO-BE CRUD Matrix

Activity	User (D1)	Child Profile (D2)	Screening (D3)	Notification (D4)	Report (D5)	Behavior Monitoring (D6)
Register User	C					
Login User	R					
Update User Profile	U					
Manage Child Profile		C,R,U,D				
Conduct M-CHAT-R Screening		R	C,R,U			
Manage Notifications				C,R,U,D		
Generate Analytics & Reports	R	R	R		C,R,U	R
Monitor Child Behavior						C,R,U,D

Table 13: CRUD Matrix

13.1.5 Physical DFD TO-BE Event Response Table

Event	Source	Trigger	Activity	Response	Destination
Parent register account	Parent	Parent submit information	Send registration details to the User data store. Send email verification.	Profile creation confirmation	Parent
Parent logs in	Parent	Parent submit login credentials	Verify login. Accept or deny access. Record login details.	Access accepted or denied.	Parent
NGO logs in	NGO	NGO submit login credentials	Verify NGO login. Accept or deny access. Record login details.	Login success or denied	NGO
Parent updates child profile	Parent	Submit child data	Store updated child profile to Child Profile data store	Update child profile	Parent, NGO
Parent completes M-CHAT-r screening test	Parent	Parent submit M-CHAT-R responses	Present M-CHAT-R question. Receive an answer. Calculate risk classification. Store answer and risk level in screening data store.	Risk level and screening outcome	Parent
Parent need notification	System	Schedule trigger	Generate daily behavior remainder. Store notification in Notification data store.	Daily routine activity remainder	Parent
NGO request report	System	NGO request report	Generate formatted report and store in Report data store	Formatted report	NGO
Parent manage child behavior	NGO	Parent submit child behavior	Update daily child behavior in Behavior Monitoring data store	Child behavior report	NGO

Table 14: Event Response Table

13.1.6 Physical DFD TO-BE Structure Chart

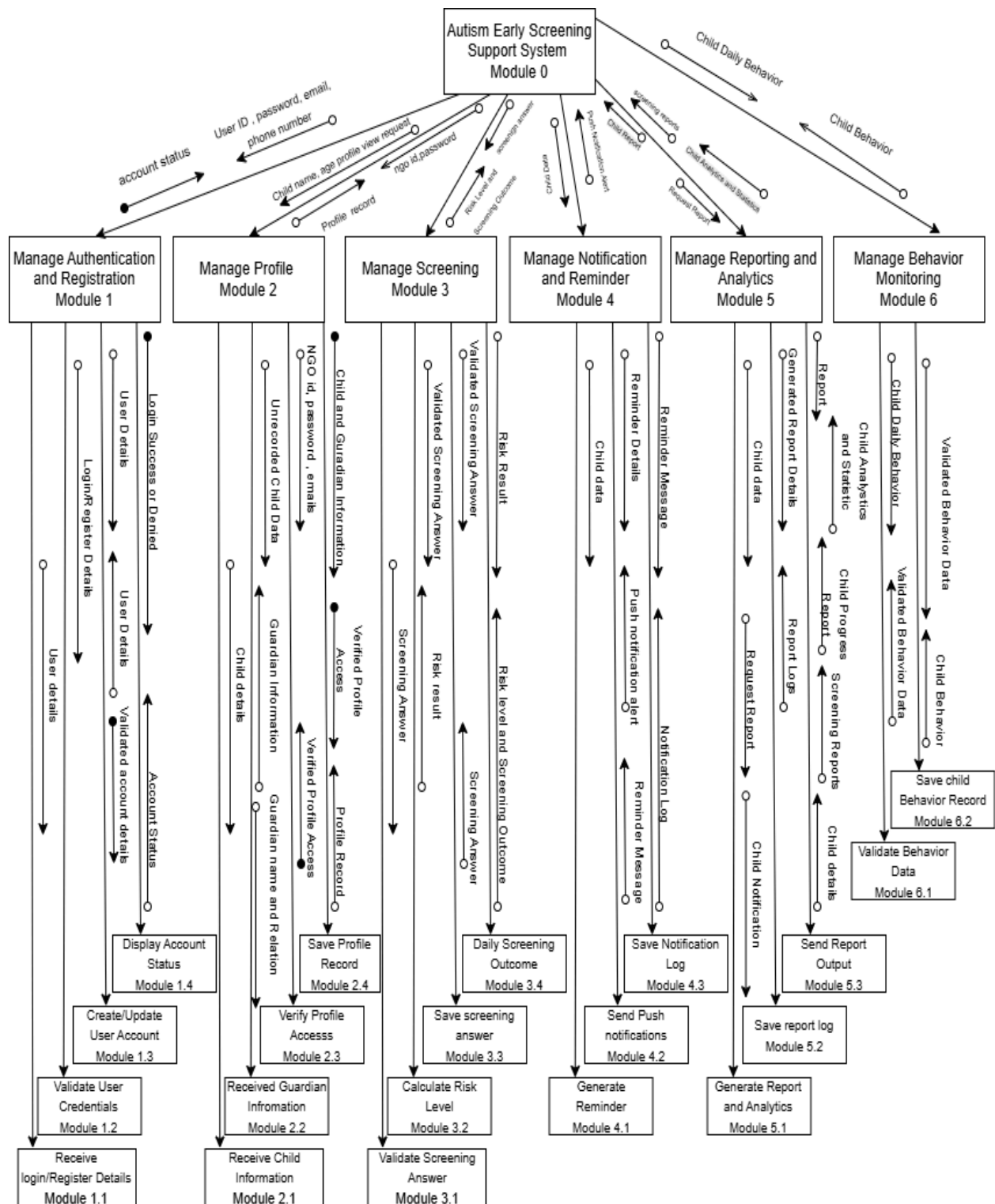
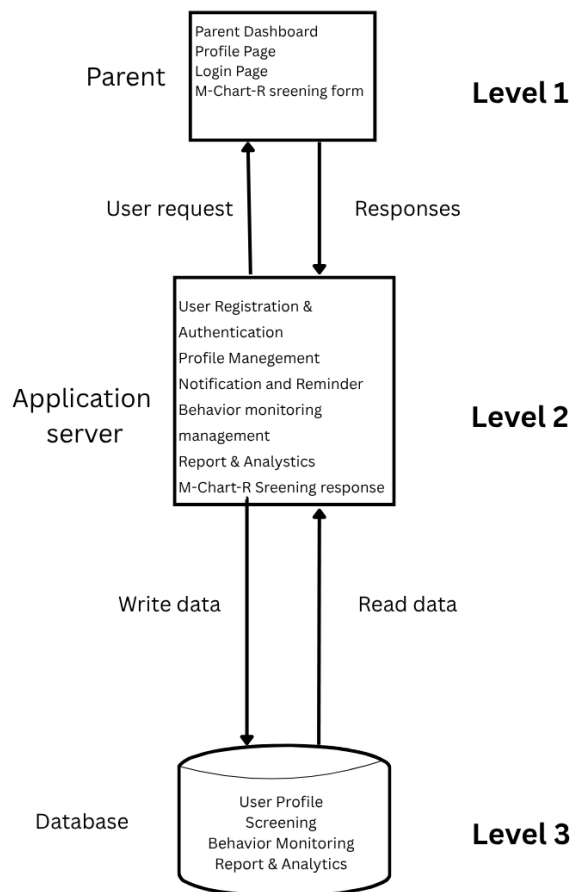


Diagram 28 : Structure Chart

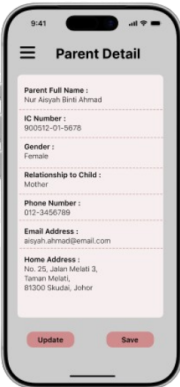
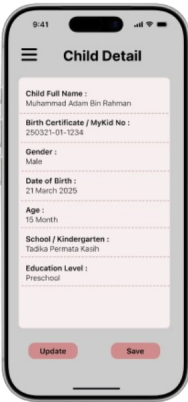
This is a comprehensive Structure Chart that maps out a proposed digital system. It divides the high-level application into six distinct functional modules, ranging from authentication and profile management to reporting and behavior monitoring based on physical DFD.

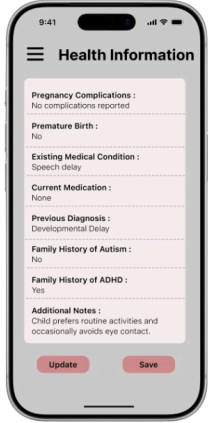
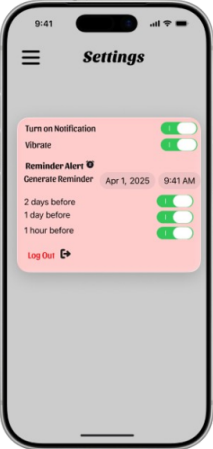
13.1.7 Physical DFD TO-BE System Architecture



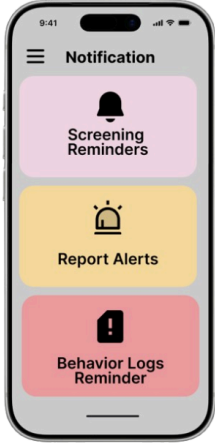
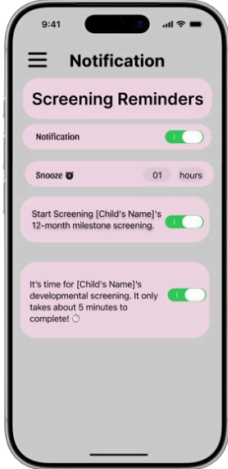
14 .0 System Wireframe

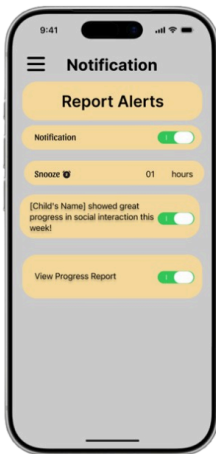
Profile

Prototype	Description
	This is the Profile screen which serves a central hub for users to manage their account and personal information. It features a user avatar placeholder at the top and the list of clear, navigation buttons allows users to easily access specific sub-sections, including "Parent Details", "Child Details", "Health Information", "BPPOKU Form", and "Settings".
	This is the section of Parent Detail screen, which displays the comprehensive personal information of the parent or guardian. The details including Parent Full Name, IC Number, Gender, Relationship to Child, Phone Number, Email Address, and Home Address are clearly organized. At the bottom of the screen, there are buttons labeled "Update" and "Save" are provided, allowing users to modify their personal records or save changes.
	This is the section of the Child Detail screen, which displays the important personal and educational information of the child. The details including Child Full Name, Birth Certificate/MyKid No, Gender, Date of Birth, Age, School/Kindergarten, and Education Level are write down. At the bottom, there are buttons labeled "Update" and "Save" are provided, allowing users to modify or save the child's profile information.

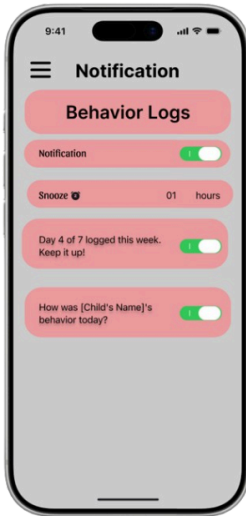
	This is the section of Health Information screen, which outlines the medical and developmental history relevant to the child's screening. The health records covering Pregnancy Complications, Premature Birth, Existing Medical Condition, Current Medication, Previous Diagnosis, Family History of Autism, Family History of ADHD, and Additional Notes are displayed in a structured card format. At the bottom, there are "Update" and "Save" buttons allow users to modify or update the health records.
	This is the section of BPPOKU Form screen, which displays a digital or embedded version of the official Malaysian registration form for persons with disabilities (OKU). The document contains detailed fields organized into distinct sections with blue headers, covering registration categories, applicant/guardian details, personal information, education, and employment history. Users can view and fill out the detailed fields directly on the screen
	This is the section of Settings screen, which allows users to configure application preferences. It includes toggle switches for basic configurations such as "Turn on Notification" and "Vibrate". Under the "Reminder Alert" section, users can set specific dates and times to generate system reminders, along with granular toggle options to receive alerts "2 days before", "1 day before", or "1 hour before" an event. A "Log Out" action with an exit icon is positioned at the bottom of the card

Notification

Prototype	Description
 A mobile app prototype of a notification screen. It features a header with a hamburger menu icon and the title 'Notification'. Below the header are three large, vertically stacked, rounded rectangular buttons. The top button is pink and contains a bell icon and the text 'Screening Reminders'. The middle button is yellow and contains an alarm icon and the text 'Report Alerts'. The bottom button is coral and contains a document icon with an exclamation mark and the text 'Behavior Logs Reminder'.	This is the Notification screen, which acts as a landing menu for managing different types of system alerts. It displays three large, color-coded block buttons stacked vertically for quick navigation. The top pink button features a bell icon for "Screening Reminders", the middle yellow button includes an alarm icon for "Report Alerts", and the bottom coral button displays a document alert icon for "Behavior Logs Reminder".
 A mobile app prototype of a 'Screening Reminders' settings screen. It has a header with a hamburger menu icon and the title 'Notification'. Below the header is a section titled 'Screening Reminders'. This section contains a 'Notification' toggle switch (turned on), a 'Snooze' field set to '01 hours', and two individual notification preview cards. The first card is for 'Start Screening (Child's Name)'s 12-month milestone screening' with an active toggle switch. The second card is for 'It's time for (Child's Name)'s developmental screening' with an active toggle switch and a small circular icon at the bottom right.	This is the section of Screening Reminders, It allows users to manage specific alert preferences for developmental testing. The configurations include a main "Notification" toggle switch, a "Snooze" adjustment field currently set to "01 hours," and two individual notification preview cards with active toggle switches for "12-month milestone screening" and general "developmental screening" alerts.

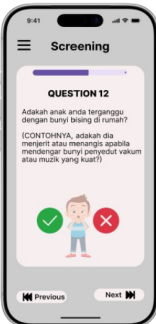


This is the section of Report Alerts,it designed for managing notifications related to the child's developmental milestones and summary data.The available options include a main "Notification" toggle switch, a "Snooze" duration input set to "01 hours," an alert preview card for weekly social interaction updates, and a dedicated toggle for "View Progress Report" notifications.



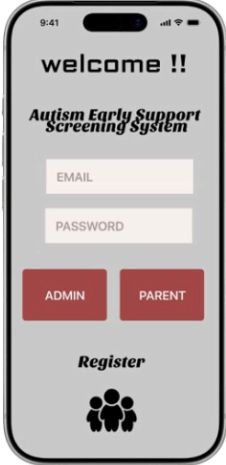
This is the section of Behavior Logs,it allows users to configure alert preferences for tracking and recording daily behavioral updates.The configuration elements include a master "Notification" toggle switch, a "Snooze" duration selector set to "01 hours," an encouraging weekly progress card ("Day 4 of 7 logged this week").

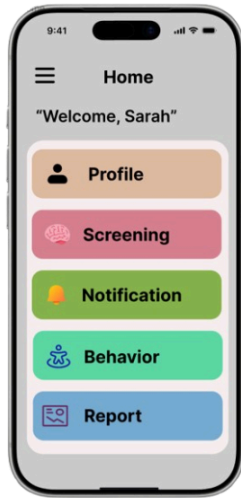
SCREENING

Prototype	Description
	This is the welcome screen that introduce the purpose of the M-CHAT-R test as a screening tool for Autism Spectrum Disorder (ASD). It will gives parents simple brief instruction before they click the "Start Test" button to start.
	This screen displays the first of the 20 questions M-CHAT-R, it will asking parents about specific toddler behaviors like pointing or looking at objects. It include clear "Yes" (green check) and "No" (red cross) buttons together with a progress bar at the top.
	This page will shows an ongoing test screen where the progress bar has advanced to Question 12, which asks about the child's reaction to loud noises. It include a "Previous" button so users can easily go back and correct a previous answer if they want.
	This results screen appears when a child scores between 0 and 2, indicating a low risk for ASD. It feature a prominent green warning triangle and advises parents that no immediate action is needed unless future concerns arise.

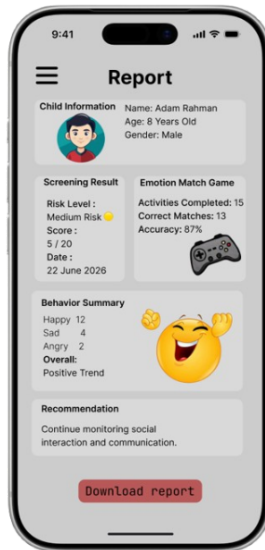
	This results screen triggers for scores between 3 and 7, indicate a moderate risk with a yellow warning triangle. It tell the user that a follow-up interview or re-screening is recommend to clarify the specific areas of concern.
	This final results screen displays a red warning triangle for high-risk scores from 8 to 20. It strongly recommend the parent to bypass further screening and directly seek a professional diagnostic to evaluate and early intervention services.

Parent

Prototype	Description
	This screen is the Parent Login Page, which restricts system access to authorized by requiring a valid identification and password that has been registered.
	This is the Welcome and Registration screen for the "Autism Early Support Screening System". It features a simple welcome heading at the top, followed by four input fields for users to fill in their details: Name, Password, Email, and Phone Number. At the bottom, a text button labeled "SUBMIT" is provided right above a family icon to complete the registration or login process.

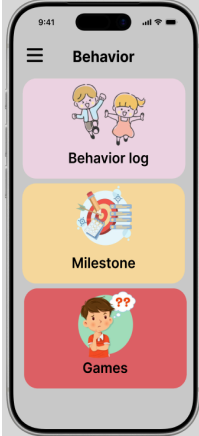
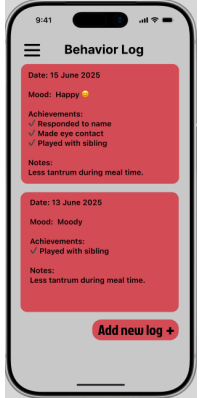


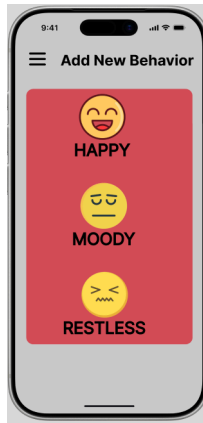
This is the Home screen of the application. It serves as the main dashboard, featuring a vertical stack of five color main menu buttons for easy navigation: "Profile", "Screening", "Notification", "Behavior", and "Report". Each button is equipped with a distinct icon representing its function.



This individual child report displays child information and data. It compiles a child's personal information, latest screening result, overall games that they played, behavior summary for current child's state and also the recommendation for guardian.

BEHAVIOR

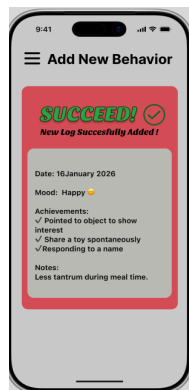
Prototype	Description
	This screen is for the behavior section for users to choose whether they want to go to either behavior log , milestone or games.
	This is a behavior log screen where it shows the past behavior log that has been filled in by the user according to the deadline. It also provides an add icon for users to input a new behavior log.
	This screen shows the first step to add a new log which is the user needs to choose the date and press the select icon to move on to the next step.



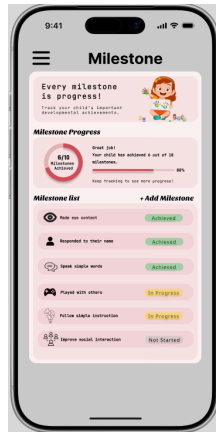
This screen shows the second step to add a new behavior log which is for the user to choose the overall mood of their child for that particular day. It provides three moods to choose , “HAPPY” or “MOODY” or “RESTLESS”.



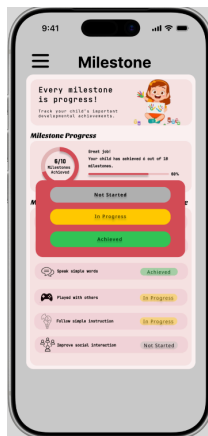
This screen shows the third step to add a new behavior log where users need to choose achievements that have been achieved by their child based on details and pictures by pressing the add icon.



This screen shows that the user has been successfully added a new log and shows the generated new behavior log details.



This screen shows the milestone layout. It shows the chart of milestone progress based on the milestone list that have been achieved. It also shows the milestone list like made eye contact and many more and the progress of it either 'achieved' or 'in progress' or 'not started'. It also has an "+ add milestone" icon for users to add a new milestone list.



This screen shows the layout when the user presses the progress button and the user can change the progress by selecting one of three progress.



This screen shows the layout when the user presses the add milestone button. The user can input a new milestone list and choose the progress.

	This screen shows the layout of the play games section. The user can press the “play” button to start the game.
	This is the game play section. The game name is “Emotion Match” where users need to choose and tap on the same face as the image provided. This layout also includes the progress of the question number at the top of the question based on the green dots. Below the question , it shows the number of correct questions from overall questions. This screen also provides the previous icon to move to the previous question and next icon to move to the next question.
	This is the layout when the play games section ended. It will show the wishes text message based on the score. In here it shows the message of “Congratulations!” and “Perfect Score”.

ADMIN

Prototype	Description																														
welcome !! Autism Early Support Screening System ADMIN_ID PASSWORD LOG IN	This screen is the Admin Login Page, which restricts system access to authorized by requiring a valid administrator identification number and password that has been registered.																														
Arifrah DASHBOARD REPORT DASHBOARD REPORT DASHBOARD Total Child 2500 Child Gender Risk Level Trend Child month	This Admin's Dashboard functions as an analytics hub, which automatically collects and visualizes system data to help the admin track all the child's data. It calculates the total number of registered children, distribution by gender, graph of risk level for each month and also a group of risk level by child's month.																														
Arifrah DASHBOARD REPORT DASHBOARD REPORT CHILD REPORT <table><thead><tr><th>Time</th><th>Child name</th><th>Risk level</th></tr></thead><tbody><tr><td>23/6/2026</td><td></td><td></td></tr><tr><td>2:34 PM</td><td>Adam Rahman</td><td>MEDIUM RISK</td></tr><tr><td>2:10 PM</td><td>Athirah</td><td>LOW RISK</td></tr><tr><td>1:47 PM</td><td>Leo</td><td>LOW RISK</td></tr><tr><td>1:23 PM</td><td>Mira</td><td>HIGH RISK</td></tr><tr><td>1:17 PM</td><td>Tiga</td><td>MEDIUM RISK</td></tr><tr><td>1:10 PM</td><td>Firzana</td><td>HIGH RISK</td></tr><tr><td>1:04 PM</td><td>Mia</td><td>LOW RISK</td></tr><tr><td>12:59 PM</td><td>Ali</td><td>LOW RISK</td></tr></tbody></table>	Time	Child name	Risk level	23/6/2026			2:34 PM	Adam Rahman	MEDIUM RISK	2:10 PM	Athirah	LOW RISK	1:47 PM	Leo	LOW RISK	1:23 PM	Mira	HIGH RISK	1:17 PM	Tiga	MEDIUM RISK	1:10 PM	Firzana	HIGH RISK	1:04 PM	Mia	LOW RISK	12:59 PM	Ali	LOW RISK	For the report part it shows all the child report list which generate real-time child updated, links to the specific child's file
Time	Child name	Risk level																													
23/6/2026																															
2:34 PM	Adam Rahman	MEDIUM RISK																													
2:10 PM	Athirah	LOW RISK																													
1:47 PM	Leo	LOW RISK																													
1:23 PM	Mira	HIGH RISK																													
1:17 PM	Tiga	MEDIUM RISK																													
1:10 PM	Firzana	HIGH RISK																													
1:04 PM	Mia	LOW RISK																													
12:59 PM	Ali	LOW RISK																													
Arifrah DASHBOARD REPORT DASHBOARD REPORT ADAM Child Information Screening result Emotion Match Game Behavior Summary Recommendation Download report	This individual child report displays child information and data. It compiles a child's personal information, latest screening result, overall games that they played, behavior summary for current child's state and also the recommendation for guardian.																														

15.0 Summary of the Proposed System

To sum up, the proposed system is a web-based platform designed to track child development milestone and provide early developmental screening. This way, it will be more easier for parents to monitor their child's daily behaviors, complete official medical screening questionnaires and all the data that has been provided can be stored safely as it does not require any physical instrument. It also gives Non-Governmental Organizations (NGO) a secure way to analyze data and monitor community-wide risk trends.

The proposed system includes five main features such as secure access and registration, automated developmental screening, daily behavior and emotion tracking, smart reminders and advanced analytics for NGOs. By using this system, Parents can safely register profiles for themselves and their children using modern, secure Firebase authentication and encryption protocols. Not only that, parents can easily complete the built-in M-CHAT-R questionnaire right inside the app. The system features a built-in scoring engine that instantly calculates scores to identify if a child is at Low, Medium, or High risk for developmental delays. More features in the app including parents can log daily behavior updates, which the system automatically transforms into easy to read progress and performance charts. To help the parents that are busy, the system provide a background automation engine tracks schedules and automatically sends push notifications to parents when it is time for a check-in or screening update. For Authorized NGO partners, they can access a reporting dashboard to generate clean, filtered reports on risk classifications, helping them offer timely support and early interventions.

This way, it will remove the family confusion around early child development as the use of manual paper tracking has been replaced with an easy to use mobile tool. It provides instant, automated guidance on whether a professional clinical check-up is recommended. This system also helps organizations by saving their time by automating data cleaning, scoring, and report generation. This allows NGOs to spot high risk cases faster and distribute care resources efficiently.

16.0 References

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3. <https://share.google/5rsnTubPFNd3ljMqA>
4. <https://marf.org.my/home-page-new/>
5. Early Autism Project Malaysia. (2026). [Testimonial by parent of 9 year old boy with Autism]. *Parents' testimonials*. <https://www.autismmalaysia.com/parents-testimonials/>
6. Jabatan Kebajikan Masyarakat & Kementerian Kesihatan Malaysia. (n.d.). *Garis panduan pendaftaran orang kurang upaya*. Kementerian Kesihatan Malaysia. https://hq.moh.gov.my/bpkk/images/Kesihatan_Orang_Kurang_Upaya/PDF/Akta_dan_Dasar/2_Garis_Panduan_Pendaftaran_Orang_Kurang_Upaya.pdf
7. Thabtah, F., & Peebles, D. (2019). Early Autism Screening: A Comprehensive Review. *International Journal of Environmental Research and Public Health*, 16(18), 3502. <https://pmc.ncbi.nlm.nih.gov/articles/PMC6765988/>

17.0 Appendix

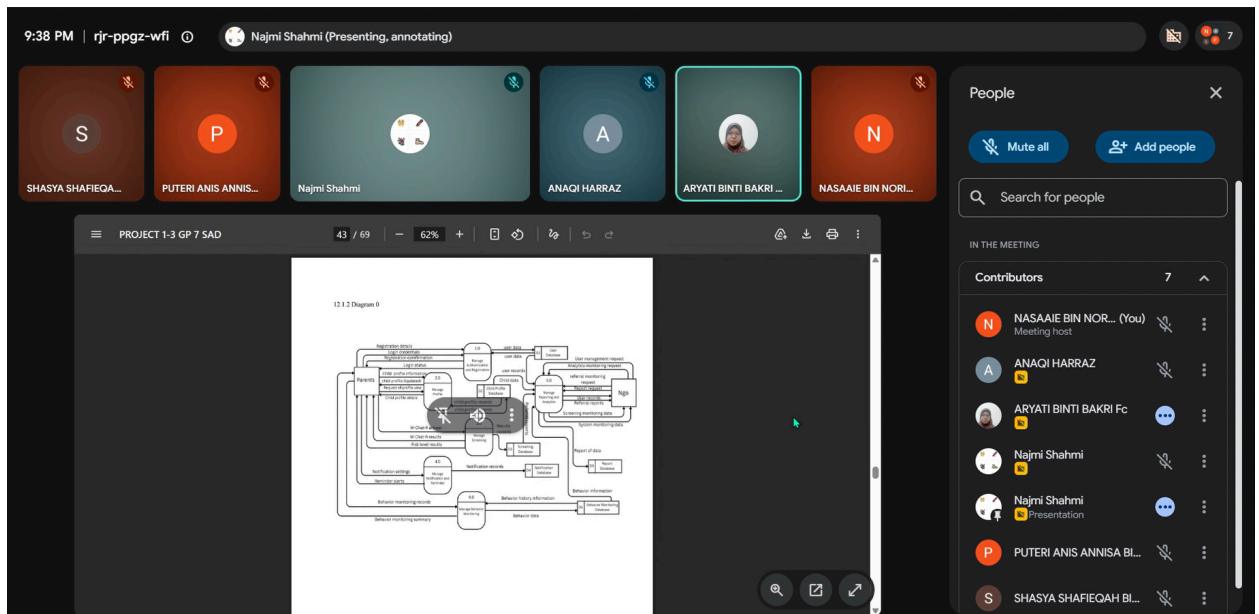
1. Google Form Link : <https://forms.gle/sjG3sWjbXcYT83ad8>

Requirements Gathering for Behavior Monitoring Module in Autism Early Screening Support System Pengumpulan Keperluan untuk Modul Pemantauan Tingkah Laku dalam Sistem Sokongan Saringan Awal Autisme

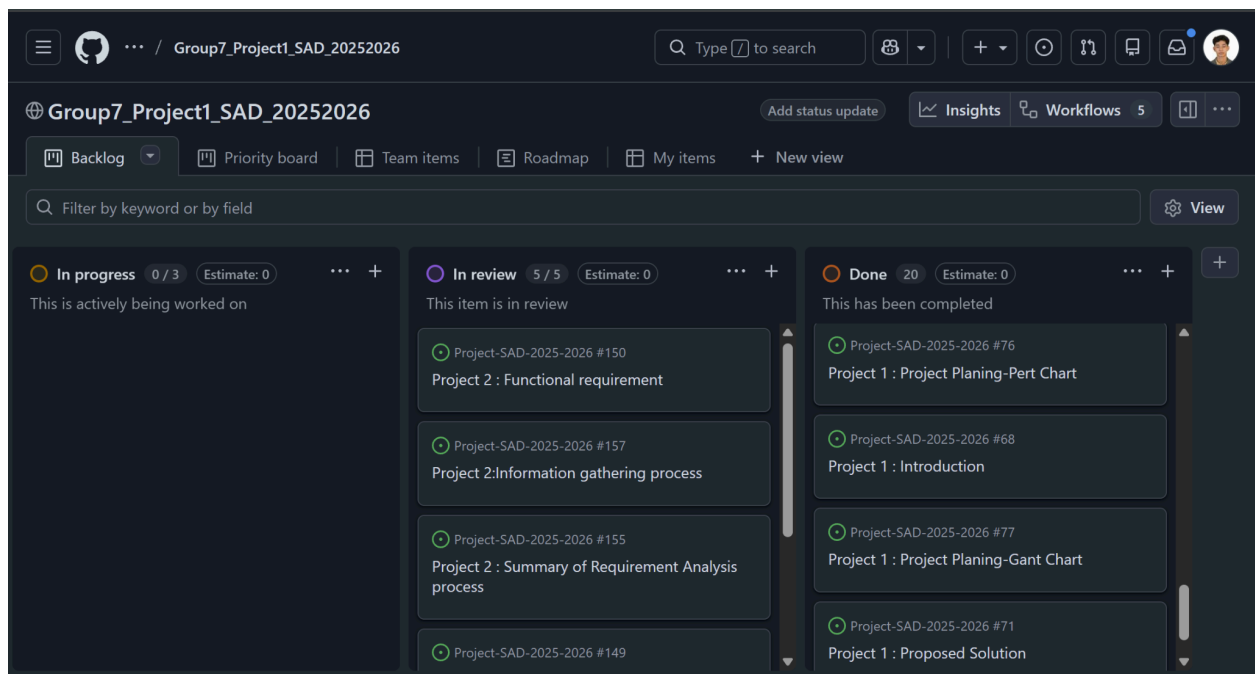
This questionnaire is conducted to gather user requirements for the Behavior Monitoring (Emotion or Mood Tracking) module in the Autism Early Screening Support System. The module is intended to help parents, teachers, therapists, and healthcare professionals continuously record and monitor the emotional and behavioral patterns of children with autism. Your feedback will help us design a practical, user-friendly, and effective system.

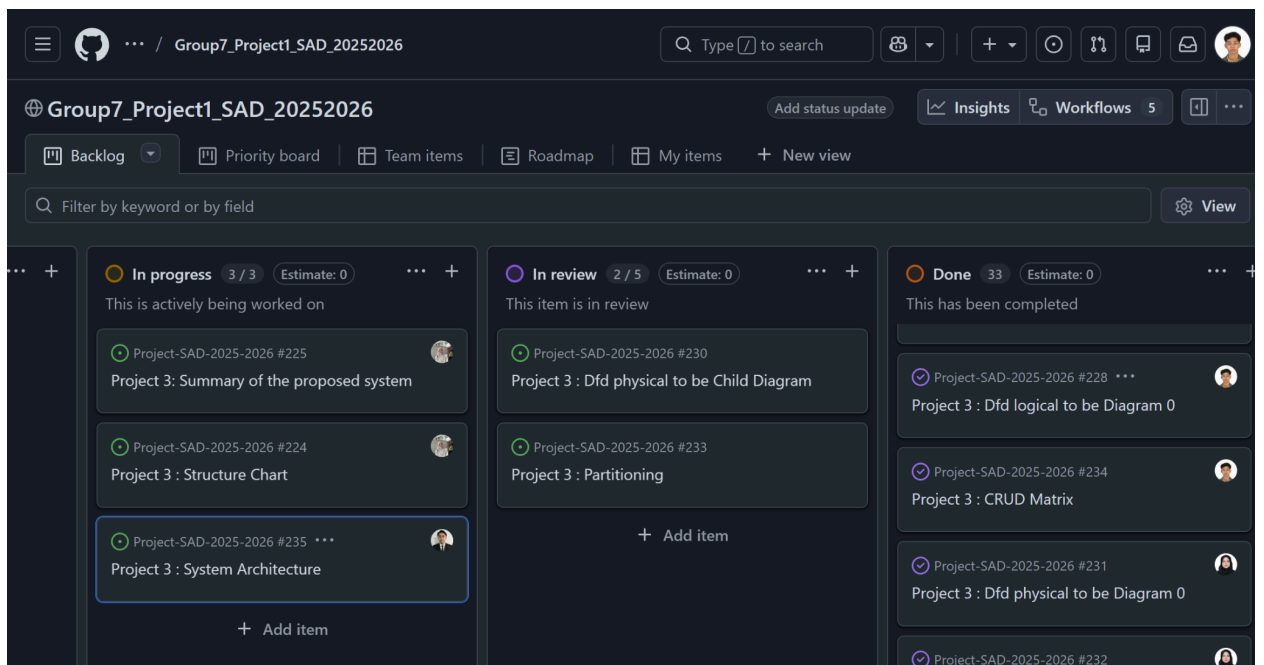
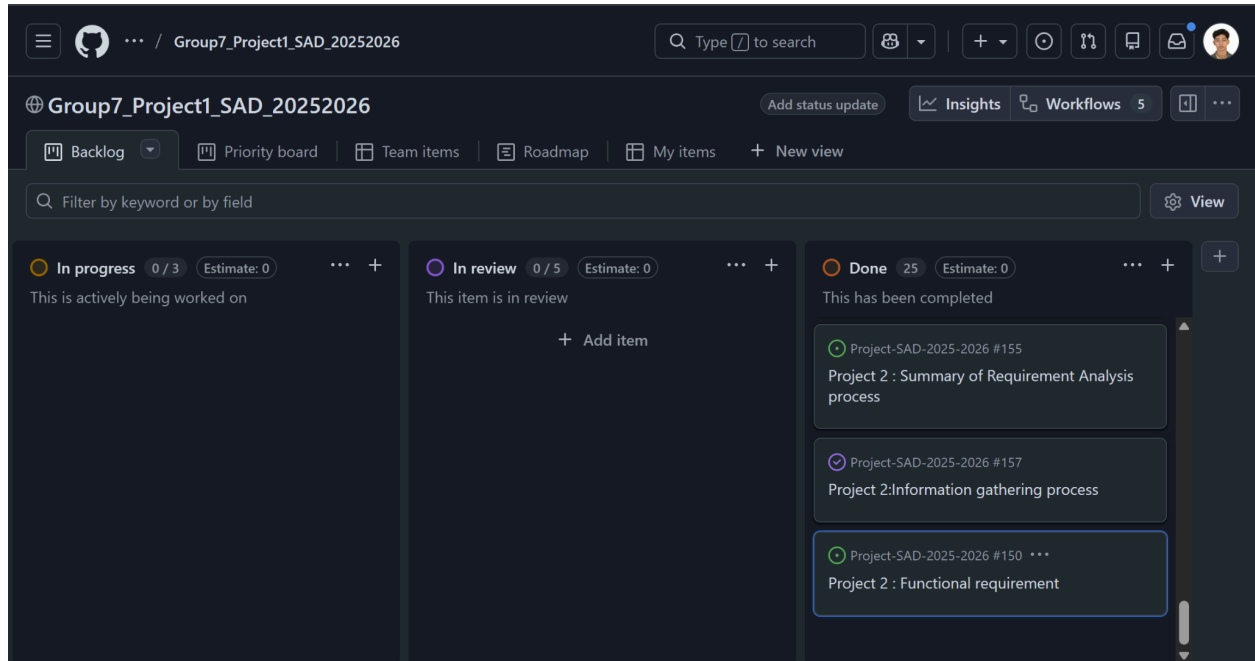
Soal selidik ini dijalankan untuk mengumpul keperluan pengguna bagi Modul Pemantauan Tingkah Laku (Penjejakan Emosi atau Mood) dalam Sistem Sokongan Saringan Awal Autisme. Maklum balas anda akan membantu kami membangunkan sistem yang praktikal, mesra pengguna, dan berkesan.

2. Group Meeting



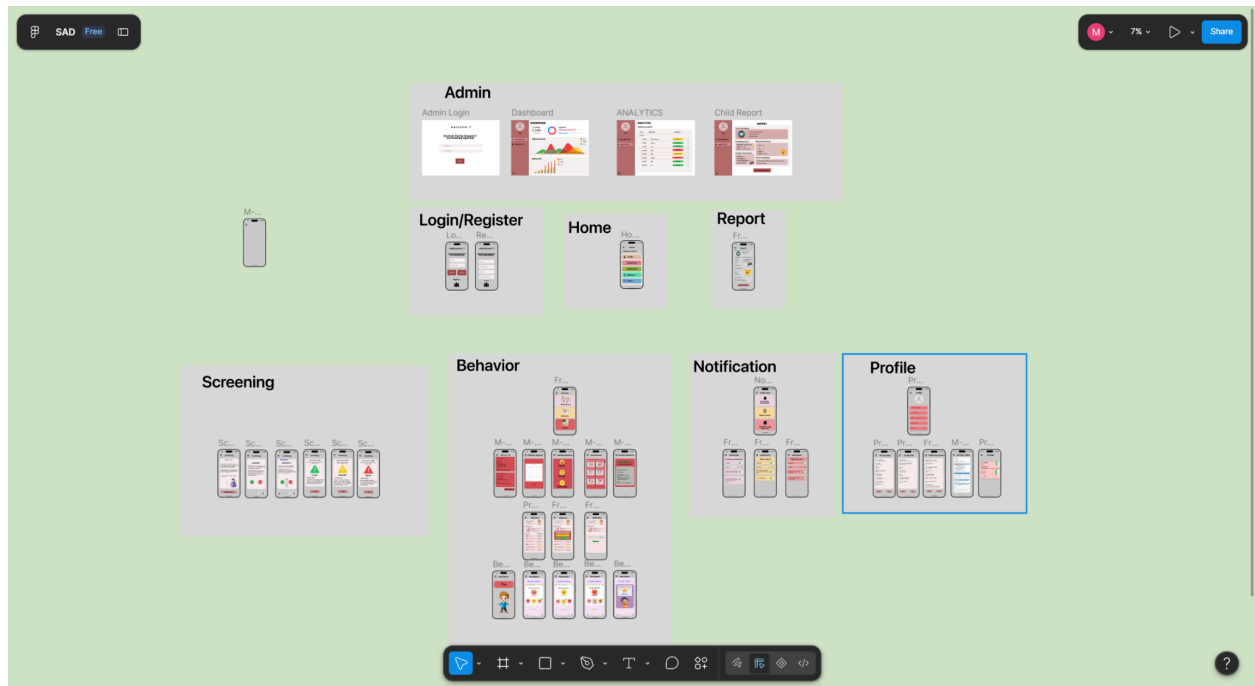
3. Backlog : <https://github.com/MFeroz06/projects/52>





4. Figma :

<https://www.figma.com/design/SUMGuSN41as9v7kUJrqlyM/SAD?node-id=0-1&t=70EvmJZxDILuRDJj-1>



5.Demo video link <https://youtu.be/ZArHwMiug2Q>